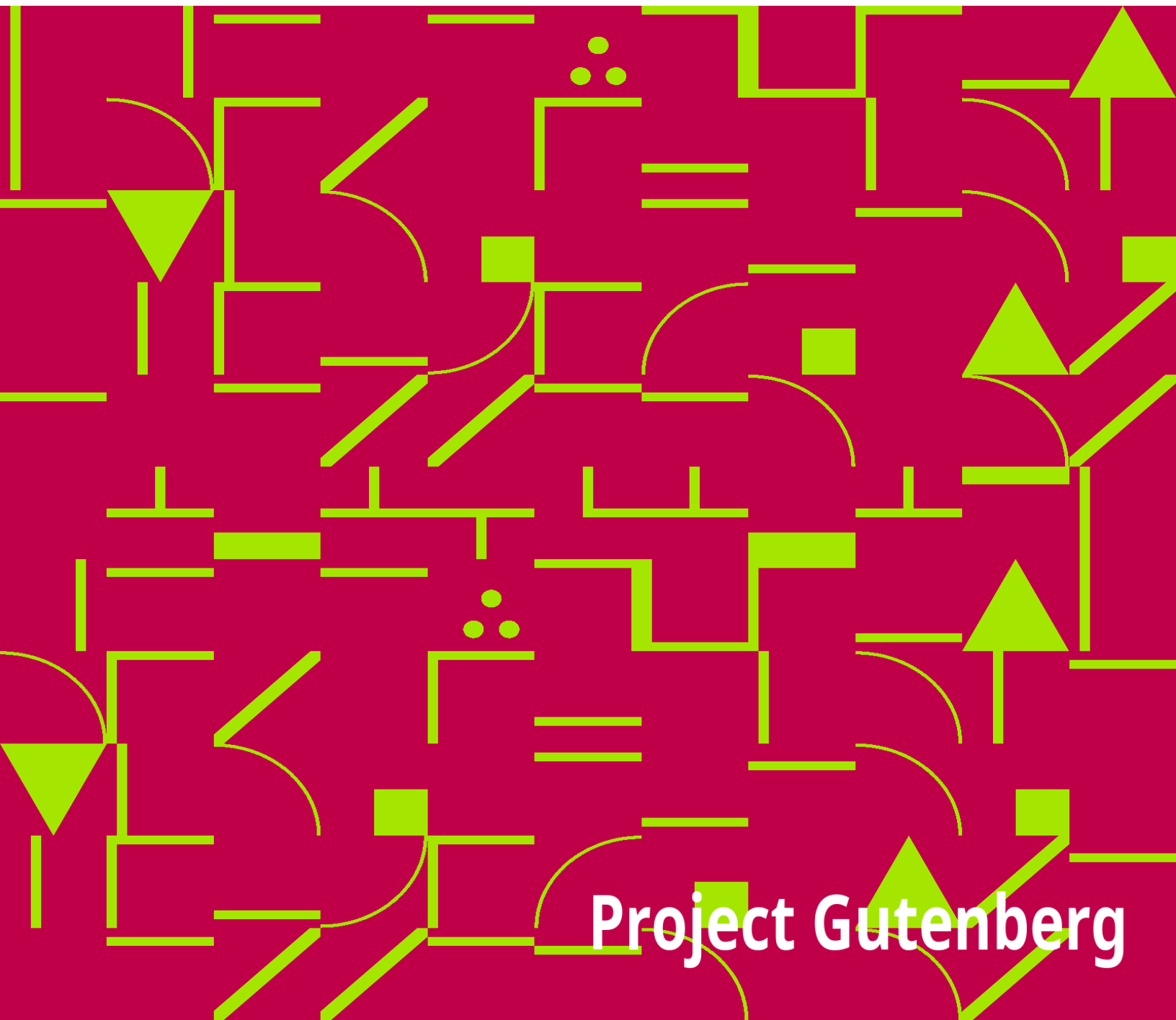


# The exploration of Tibesti, Erdi, Borkou, and Ennedi in 1912-1917

Jean Tilho



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Author: Jean Tilho

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EXPLORATION OF TIBESTI, ERDI, BORKOU, AND ENNEDI IN 1912-  
1917 \*\*\*

THE EXPLORATION OF TIBESTI, ERDI, BORKOU, AND ENNEDI IN  
1912-1917: A Mission entrusted to the Author by the French Institute

Lieut.-Colonel Jean Tilho, Gold Medallist of the R.G.S. 1919

*Read at the Meeting of the Society, 19 January 1920. [Map](#) following p. 160.*

*[Note: The names in the text are spelled in accordance with the manuscript of Colonel Tilho, a few of the principal names—as Chad—in their English form, but the greater number in the French transliteration of Arabic. On the accompanying map the names are transliterated according to the G.S.G.S. rules for transposing from the French to the British system. The retention of the French spelling in the text has the double advantage of familiarizing the student with the two systems, and of preserving in some degree the character of the lecture, which was delivered in French.—ED. G.J.]*

## 1. Object of the Mission.

**B**EFORE I begin my lecture, allow me to express once more, in your presence, my heartfelt gratitude to the Council of the Royal Geographical Society for the high recompense accorded me on the occasion of my last journey in Central Africa.

It is of this journey, its chief incidents, and most important results, that I am about to have the honour of giving some account. Let me first of all explain to you, in a few words, what, from a geographical point of view, was the object of my expedition.

Explorations in Central Africa, made during the second half of the nineteenth century and in the beginning of the twentieth, had left unsolved a very interesting problem: it had been noticed that the level of vast stretches of desert, several hundred miles north-east of Lake Chad, were considerably lower than that of the lake—the difference amounting in some places to 260 feet; besides this, a wide continuous trench, offering the appearance of an old valley—the Bahr El Ghazal—led from the lake to this low-lying ground, and seemed to stretch far away to the north-east, between the mountain groups of Tibesti and Ennedi. On proceeding towards the north-east, an increasing analogy is to be noticed between the malacological fauna of the Chad basin and that of the Nile. Besides which there had been found recently, in the waters of the Chad, a shrimp till then only found in the Nile basin—the *Palæmon Niloticus*, Roux. In short, all these signs appeared to confirm the supposition that the basin of the Chad was not a closed basin, but belonged to that of the Nile, and was a former affluent of the old river on whose banks had sprung up and flourished one of the most brilliant and ancient civilizations of the world.

This was the hypothesis that the French Institute wished to have investigated, and in the early part of 1912 I had the honour to be chosen to undertake the necessary researches. May I tell you how the mission thus entrusted to me fulfilled my dearest wish? From my early youth I had felt myself irresistibly drawn towards Africa, and I was filled with a desire to

take a modest share in the discoveries of great explorers, whose intrepid expeditions had revealed to the civilized world some part of the mysterious and immense dark continent.

You doubtless remember how vague, some thirty years ago, was our knowledge of that part of the world. At that time—which now seems so far away even for those then living—I had for chaplain at the grammar-school a holy man who was an ardent patriot; in his Sunday sermons he used to talk to us a little of our duty to God, and still more of our duty to our humiliated country, which was waiting and meditating, as it laboured, on the possible reparation of the iniquities of 1871. His voice, sad at first while he spoke of our disasters and the sufferings of our lost provinces, soon grew eager and thrilled as he showed us the new way to be taken by children, as we then were, to raise the prestige of our flag: he would speak to us of that mysterious Africa, half revealed by Livingstone, Stanley, and Savorgnan de Brazza; and I fancy, after these thirty years, I still hear the sound of the name of Savorgnan de Brazza re-echoing through our humble chapel and thrilling like a bugle-call. Then, of an evening in the class-room, I would ponder over the map of Africa, where amid great blank spaces appeared in the centre of the continent a few geographical features, one of which, coloured in blue, Lake Chad, possessed a singular fascination for me.

Some years later, on leaving Saint-Cyr, I began to look forward to the realizing of my dream: after a first campaign in Madagascar, I was sent out to serve on the banks of the Niger in 1899; and since that date each successive campaign in Africa allowed me to push a little further eastwards, and so get to work on a fresh item of the programme I had set myself to carry out: to establish an accurate geographical liaison between the basins of the Niger, the Chad, and the Nile, and unite by a great transversal line the extreme ends of the routes followed by Nachtigal to Tibesti, Borkou, Wadai and Dar Four.

In 1912 I was ordered to take command of the province of Kanem for the purpose of preparing a projected expedition against Borkou, where the Senoussists had established their chief centre of agitation and anti-French propaganda, and whence they periodically sent out plundering expeditions, which spread ruin and desolation among the peaceful tribes placed under

our protection. About the same time, the Académie des Inscriptions et Belles-Lettres entrusted me with the mission I mentioned above, concerning the supposed connection between the basins of the Chad and of the Nile. Of this latter expedition, which lasted five years—1912-1917—I now propose to give you a *résumé*.

## 2. From Congo to Borkou.

*From Congo to Lake Chad.*—I do not think there would be any real interest in a detailed account of my journey to Kanem; I followed a route pretty well known, the Congo-Ubangi-Shari route. We left the steamer at Matadi, at the foot of the cataracts, and took the Belgian railway which leads to Kinshassa on Stanley Pool, at the head of the cataracts; from there, after crossing the Congo to land at Brazzaville, we proceeded on a river-steamer, first up the Congo itself, and then up its tributary the Ubangi, as far as Bangui. Farther up, lighter steamers enabled us to surmount the rapids and reach Fort De Possel, a little post built on the right bank at the point where the Ubangi changes its course. From Fort De Possel we went by land to Fort Crampel, covering nearly 160 miles of the zone which divides the waters between the basins of the Congo and the Chad. A fine road for motor-cars was being completed when I passed, but the only means of transport was carriers on foot. At Fort Crampel we embarked in small boats and descended the Gribingui till it falls into the Bahr-Sara, taking farther down the name of Shari; from thence we proceeded on a river-steamer up the Shari till we reached the Chad, and crossed over to the post of Bol, on the northern shore of the lake; and finally, in four more stages, we reached by land the town of Mao, the military and political centre of Kanem.

This journey, which takes about twelve or fourteen weeks, according to the season, is very interesting for travellers, and especially for sportsmen, who find opportunity for exercising their skill on game of all sizes, from the elephant and the lion to the modest guinea-fowl. I may mention that when I passed by the banks of the Shari, the remembrance of the exciting hunts of the celebrated aviator Latham, killed by a buffalo, was still fresh in every one's mind; but does any one remember Latham now? We should notice that this line is still far from comfortable, and that the ever-present danger of catching the sleeping sickness through the myriads of glossina-flies that may sting the traveller, spoils all the pleasure one would feel in beholding the splendid landscapes of tropical rivers flowing beneath the shady arches of the quiet forests.

*A Year in Kanem* (1912-1913).—I will pass briefly over the twelve months' period of my command in Kanem and the neighbouring districts. My daily task—military, political, administrative, and judicial as well—was such that the days seemed too short for the business to be done. It must be said indeed that the Kanembus, the Budumas, the Toubous, and the Arabs of this region may be reckoned among the most quarrelsome and litigious people one can imagine.

But the great matter was to be informed in time of the Senoussist raids, and when that could not be done, to discover and cut off their retreat towards their distant haunts; but we had to do with old stagers of the Sahara, who knew admirably well to wait for the right moment, and beat a rapid retreat with their booty once the thing was done.

Another important matter was the material preparation for the expedition planned against Borkou and Tibesti, where the Senoussists assembled their bands of brigands, and where they concealed their booty: camels, horses, cattle, and, above all, women and children, carried off into slavery.

The secrecy of this expedition was ensured through the simple fact that our enemies' spies had so often announced the formation and imminent setting out of a punitive column, as to render the Borkou gentlemen quite incredulous of its possibility; they were startled, however, when in July I led a reconnoitring party to the extreme limits of our frontier, but as I retraced my steps without going beyond this line, they were confirmed in their opinion that we should not dare to attack their fortress of Ain Galakka, and they recommenced more boldly than ever their incursions and plunderings among our villages and our tribes. For this reason, when, in the early November of 1912, Colonel Largeau came and assumed the command of an expeditionary column, our departure for the north-east was not considered by the Senoussists of Borkou as more threatening to them than any reconnoitring party of the preceding months had proved to be.

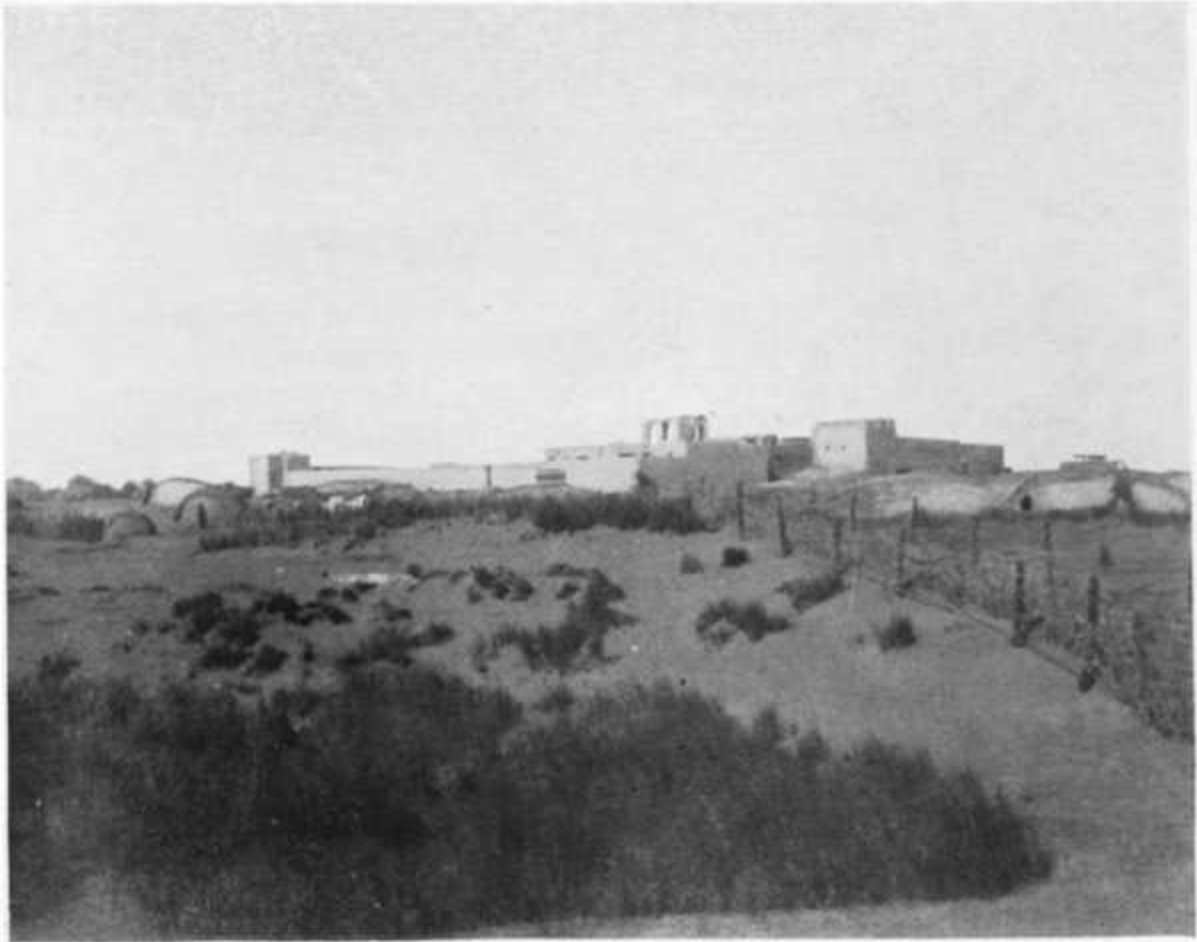
### 3. In Borkou.

*The Conquest of Borkou.*—Our expedition consisted of 400 black soldiers, with two mountain-guns; about 200 Arab and Toubou volunteers, forming a “goum” or party of scouts, accompanied the column. We carried with us provisions for forty days, and the total number of our camels was about 2000. By a rather extraordinary piece of good luck, our forward march was not disturbed by the enemy. The season was favourable, the days not being over-hot, and the nights fairly cool; the usual temperature at sunrise was about 60° Fahr., but a very strong wind, blowing from the north-east and raising blinding clouds of sand, made it seem a great deal colder. Our march was skilfully concealed as far as Kourouadi, a point from which we could threaten the fortress of Ain Galakka as easily as that of Faya. There, after allowing the troops a day for rest and final preparation, it was decided to strike a decisive blow at Ain Galakka, the principal centre of the Senoussist forces.

Our column, leaving its convoy a dozen kilometres in the rear, under a guard of fifty men, appeared before Ain Galakka on the morning of 27 November 1913; the enemy were completely surprised. The attack began by a bombardment of no more than about a hundred shells, which did great damage inside the *zawia*, and made in the outer wall many a breach for the infantry to pass through. The assault was opened at ten o'clock; the defenders, though not numerous, offered a vigorous resistance, preferring to die rather than surrender; by mid-day the entire fortress was in our hands. We had about forty casualties, of which a third were killed.



THE COLUMN HALTED AT THE WELLS OF KOUROUADI, BORKOU



THE FORT OF BERRIER-FONTAINE, OASIS OF FAYA



ROCKY COUNTRY BETWEEN THE OASES OF YARDA AND BÉDO, BORKOU



DANCE OF THE NAKAZZAS, OASIS OF FAYA, BORKOU

Leaving our wounded in Ain Galakka with a small garrison, we marched on the *zawia* of Faya, which we entered without striking a blow on December 1. Thence proceeding still farther into the desert, we reached in a week's time Gouro, a point 200 kilometres north, the religious and political centre of the Senoussists in Central Africa, which was seized after a short struggle. Then, continuing its successful march towards the east, the column took possession unopposed of the oasis of Ounianga, 60 miles from Gouro, and leaving a small garrison there we returned to Faya, the best place to be chosen for the military and political centre of the newly conquered territory.

*Importance of the Conquest of Borkou.*—This laborious campaign had the very important result of depriving the Senoussists of the valuable *tête de pont* on the south side of the Sahara which Borkou constituted for them,

enabling them to distribute over Central Africa arms, ammunition, and propagandists of the holy war.

The great value of our conquest appeared plainly a few months later, when the German Emperor let loose on the world the most awful war that ever convulsed the Universe: a Germano-Turkish mission, headed by Nuri Bey, a brother of Enver Pasha, the Turkish Minister of War, landed in Cyrenaica for the purpose of organizing, with the help of the Senoussists, an outbreak in Central Africa against the protectorates of France and Great Britain. This would have been an easy matter if our enemies had been able to establish their headquarters in Borkou, for they would then have been only a few hundred miles from German Bornou on one side, and on another from Dar Four and Dar Sula, which showed a certain hostility towards us. There is no doubt that, in this case, the Anglo-French campaign in the Cameroons would have been conducted in very different circumstances; when we take into consideration the large stock of arms and ammunition prepared by the Germans in their colony, and the care they had taken to fortify the mountain of Mora, we may suppose that the German staff had hoped to establish by main force a continental junction between the Cameroons and Turkey, through Kanem, Borkou, and Libya, in case of the communication by sea being cut off. And I do not think I shall betray any State secret by informing you that the Chad territory, with its modest resources in men and ammunition, would have been very difficult to defend with any chance of success against such an attack. I may also add that, had the Turco-Germans been able to accomplish their design, the result would have been exceedingly perilous for Franco-British rule throughout the whole of Dark Africa.

By uniting, under my command, our frontier territories of the Libyan desert, the French Government's aim was to constitute a force able to resist any attempts that might be made to retake from us the excellent base of operations that Borkou afforded.

*Four Years in Borkou* (1913-1917).—I do not think it would be of any great interest to lengthen this geographical lecture by explaining to you the difficulties of every kind that I was obliged to overcome during about four consecutive years, in order to fulfil the military task allotted to me. As

Borkou produces little else but dates, and Ennedi scarcely anything at all, I was compelled to procure from Kanem and from Wadaï the corn, meat, and other food-stuffs necessary for the maintenance of my civil and military subordinates. Now, the organizing of the commissariat transport became more and more difficult every six months; the want of pasture along the roads we had to take, the incessant raids of the nomads and the counter-raids of my troops, caused irreparable losses among our camels. From the end of 1913 to the first months of 1917, the activity of the rebels was so great, owing to the instigation of the Turco-Senoussists, that my troops could get no rest.

*A Bird's-eye View of the Country.*—When on leaving the shores of Lake Chad we proceeded towards the north-east, we first entered into a sandy region, with parallel valleys running between grassy downs that rose to a height of not more than 300 feet: this was Kanem, the country of corn and cattle, where subterranean water abounds and where it is easy to live.

After marching for about 100 miles, we left this fertile country and dropped quite suddenly into the desert itself, with its dull, empty, vague horizons, so monotonous that the slightest details interested us, such as a line of stones on the sand, the sight of a crescent of sand-dunes, or a poor, solitary, half-dead shrub; also our passing through a meagre pasturage of dusty *had* was quite an event, or the discovery in the distance of a few green bushes of *siwak*, till we reached the wells, where we were to rest all day long, to lead the camels to drink, and renew our own provision of water, which was often brackish and evil-smelling. This was the deceptive desert of the Lowlands of the Chad, the region I mentioned above as being lower in level than Lake Chad itself.

After a further march of about 250 miles we entered the country of rocks; at first scarcely visible above the sands, they soon rose in sharp peaks that looked like mediæval ruins, and then shot up into long steep cliffs bordering rugged plateaux, that formed ledges one above the other to the foot of the mountains: this was the region of Borkou, Tibesti, and Ennedi, the very heart of the desert, situated at almost the same distance from the shores of Lake Chad, the Nile, and the Mediterranean. This rocky belt forms, from the Tripolitain to Dar Four, a long broken wall, encircling

on the north-east the basin of the Chad, which it divides from the dismal and unexplored waste of the Lybian Desert. Tibesti and Ennedi form the highest and almost inaccessible parts of this region, while another part, Borkou, consists of a wide depression between the basins of the Chad and of the Nile.

#### 4. The Oasis of Borkou.

*Faya*.—The *zawia* of Faya had been chosen as the military and administrative centre of French Borkou, in preference to those of the Senoussists (Ain Galakka and Gouro), because it offers the least unfavourable lines of communication with the garrisons of Gouro, Fada, and Ounianga, and the best position for joining Borkou by wireless telegraphy to the nearest post of the Chad territory, 350 miles to the south.

The huts of the Senoussist *zawia* sheltered us from the sun and the sand-storms, but they were in such a state of ruin and decay that we were obliged to begin at once and make bricks—unbaked, of course. Unluckily, for constructing our buildings we were obliged to depend on the work of the few black soldiers who were not employed in exterior operations; so that many months elapsed before we could build a sufficient number of habitable houses, and complete the detached works of our defensive arrangements, including three rows of rope network, supposed to be barbed, by means of the addition of long thorns from the date-trees.

The landscape from the summit of the square donjon which overtopped the fort, though wanting in charm and beauty, was not without a style of its own; the post was built in the middle of a broad valley, closed in on the east, but opening spaciouly towards the west; its rugged, steep, rocky sides plunging into shifting sands and wind-swept dunes, each dune curved into the form of a crescent.

At the foot of the fort the axis of the valley was delineated by fine rows of date-bearing palms, about 500 yards wide by 20,000 long, broken at intervals by heaps of moving dunes. On either side of the palm-grove there stretched green meadows, which looked as though they would afford fine pasturage for cattle, but which in reality were covered with sharp, hard grasses and herbs of no nutritive value: the most characteristic and the least bad was *akul*, a regular little bush of sharp thorns, which the camels would eat, but not without making a funny grimace at every mouthful.

All along the valley there lies a sheet of subterranean water, which rises in some places so near to the surface that the gazelles and jackals easily slake their thirst by scraping away with their feet a few inches of the soil; here and there, indeed, a little stream of water flows out of the sand, and runs a few yards towards a neighbouring depression, and little pools are formed in natural or artificial hollows made in the soil.

These jackals and gazelles are the only wild animals found in Borkou; the latter are quite unapproachable by hunters, while the former remain hidden in the daytime, but come in bands at night, yelping round the villages, and penetrate boldly into inhabited enclosures to seek their prey. So cunning are they that they avoid the most ingenious traps the natives can set. The lion, the panther, the hyena, and the wild boar never pass beyond the desert boundaries of Kanem and Wadaï; even the antelope and the ostrich, though bearing thirst so well, cannot venture so far into the Sahara.

The winged domestic tribe is seen among the villages in the shape of rare squads of lean fowls; and flights of turtledoves and pigeons roost in the palm trees. A graceful species of sparrow, with black plumage and white tails, fly in and out of the rocks, and even come into our clayhouses; they sing like nightingales when building their nests, and chirp like sparrows while they watch their young beginning to fly. All round the inhabited houses the black crows may be heard croaking: they are extremely audacious, whether attempting to snatch pieces of meat roasting before a kitchen fire, or settling on the back of a wounded camel and tearing off with their beaks morsels of bleeding flesh.

Snakes are fairly common, the largest being hardly more than a yard in length and one or two inches thick; the most dangerous is the short bulky viper that lies hidden in clumps of grass, and whose bite is fatal even to camels. Scorpions abound, generally of a greenish hue, sometimes black; their sting is very painful, and may be eventually mortal to women and children.

Amidst the rocks one may find a curious eatable lizard, the “dundou”; it is inoffensive, but when it does bite, it bites so fiercely that the only way of making it let go is to pinch its tail sharply, either with pincers or with one’s teeth.

There are very few domestic animals save the ass and the goat; but small herds of oxen manage to cross the desert from November to February, when cool days, pools remaining from the rainy season, and the scanty pasturages of grasses produced here and there by the few summer showers allow them to pursue their march by short stages.

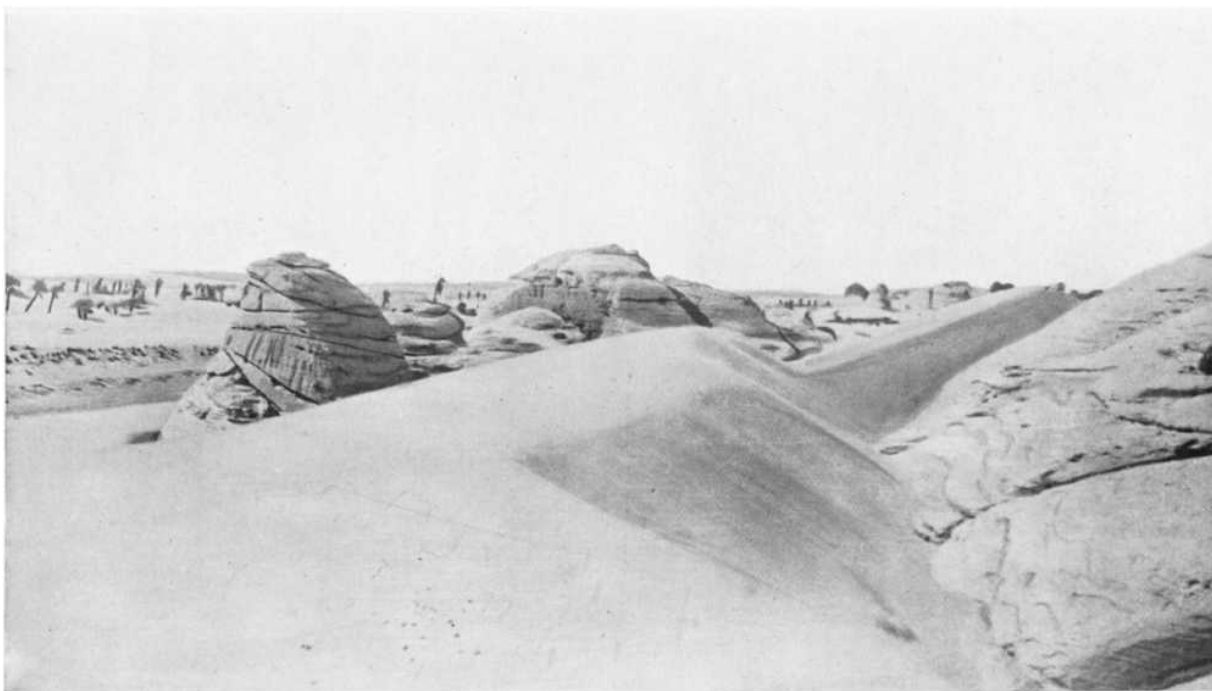
Where the animal kingdom exhibits its greatest vitality, however, is in the insect world: the common fly, dirty and worrying, rules despotically by day, together with gad-flies and big stinging flies of a pretty greenish hue. At nightfall, the very time when one might enjoy a little rest on the terrace of the houses, moths, coleopters, locusts, dragonflies, and bugs become very lively, and whirl madly round the table where a light is shining, so that it is far preferable to dine lighted only by the moon and the stars. When there is no wind at night there are swarms of mosquitoes, and also of a kind of little sand-fly that pass between the meshes of the best mosquito-nets.



SANDSTONE ROCKS NEAR ORORI, BORKOU



ROCK DRAWINGS, OASIS OF YARDA, BORKOU



SANDSTONE ROCKS ATTACKED BY MOVING DUNES, OASIS OF YARDA, BORKOU

*Cultivation.*—The soil indeed is not very fertile, which is the reverse of the account given of most oases in the north of the Sahara. It is especially favourable to the cultivation of the date-bearing palm, which loves to have its foot in the water and its summit in the burning sun, but does not stand rain well. The first dates ripen in the month of May, while the latest are gathered in September; they vary in size, and are dark or light in colour according to their variety, but nearly all are of a very good quality, as sweet and fleshy as one could wish. The greater part of the crop is put to dry, while the most luscious are gathered into heaps and pressed into goatskins, to be carried to Wadai and Kanem and other places farther off.

After the date-gathering the natives prepare their gardens for the sowing of corn, which takes place in November and December. The ground is arranged in small squares, ingeniously adapted for irrigation; but the produce is meagre owing to the want of manure; this is remedied, to a certain extent, by an addition of virgin soil, containing more or less soda, which is fetched from some distance on donkey-back. The gardens are intersected with long parallel hedges, which shelter the ears from the withering violence of the north-east wind. The harvest is gathered in towards the end of March, and a short time later the ground is prepared for

the sowing of millet, which yields a still smaller crop than the corn. When we add that in some gardens there grow a few onions and tomatoes, as well as a kind of spinach, scarcely appreciated anywhere but in Borkou, we shall have enumerated nearly all the available food-stuffs of the oases.

I must not forget to mention that the Senoussists had succeeded in importing to Gouro and Faya some fig-trees and a few vines; and on our side we managed to acclimatize the sweet potato, a precious resource which came from Kanem. We were less fortunate in our repeated attempts to acclimatize French vegetables, which succeed so well in the neighbourhood of Lake Chad during the cool season; the poverty of the soil, the want of manure, the extreme dryness of the north-east wind, the voracity of the grasshoppers and other destructive insects, were no doubt the causes of our lamentable failure as agriculturists.

*Winds and Rain.*—In the heart of the Sahara, where rain is so rare a meteorological phenomenon, the wind is the high arbiter of each day's weather. The weather is fine when the wind is light, and bad when it is strong; in the latter case nothing is to be seen but whirling columns of sand, raised by the north-east wind, blowing in stormy gusts and covering the whole landscape with a thick dry mist of brownish dust that penetrates everywhere and is very painful to the eyes, so that one does well on such occasions to wear motor-goggles to avoid ophthalmia. These north-east winds blow more or less violently for a great part of the year, sometimes for a few hours only each morning, sometimes for whole days and nights. I may say that we were able to note a fair correlation between the oscillations of the curves of the registering barometer and thermometer and the force and duration of these winds; they usually coincide with low temperatures and high atmospheric pressure, while the light winds or the dead calm accompany low pressure and high temperatures. Taking as a basis the information furnished by the natives, borne out by our four years of regular observations, it may be said that, as a general rule, the north-east wind reigns supreme over Borkou and the neighbouring districts from October to May or June (that is to say, from about the autumnal equinox to the summer solstice); whereas in July, August, and September still weather prevails, alternating with gentle west-south-westerly winds.

It is these latter winds that bring with them from the Atlantic what little moisture nature measures out each year so parsimoniously to these dried-up lands. Then the sky clouds over almost every afternoon, but one's hope of refreshing showers is vain; the heat thrown up from the scorched ground, and the rapidly rising temperature through which the raindrops fall towards the earth (a rise of about 3° Fahr. per 1000 feet), are enough to bring about their more or less complete evaporation before they reach the ground, and one sees long frayed streaks of grey cloud trailing almost along the ground, like unravelled skeins of wool, from which a few rare drops fall on the thirsty earth. When we took possession of Borkou the inhabitants assured us with one voice that it had not rained in their country for eleven years, thus putting back the date of the last rain to the year 1902; by a curious chance our entry into Faya (on 1 December 1913) was greeted by a little shower of utterly unlooked-for rain. The inhabitants saw in this downfall (unusual not only for that region, but for that season of the year) a happy omen for the rainy season of 1914, an omen which was realized, for in the month of August 1914 we had the satisfaction of registering about 90 mm. of rain at Faya. In 1915 the rainfall was hardly worth mentioning, and in 1916 about 35 mm.

Though Borkou is more than 300 miles south of the Tropic of Cancer, and very low-lying (650 feet above sea-level), the heat is really excessive only for six or seven months of the year, from mid-March to mid-October. During our observations, extending over three years, the maxima registered in the hot season never exceeded 117° Fahr., but temperatures of 110° to 115° were frequent. During the cool season, from December to February, the minima sometimes fall below 50° Fahr. without ever getting down to freezing-point. The dryness of the air is very noticeable from November to June, when a difference of more than two to one may regularly be observed between the simultaneous indications of the dry and wet thermometers: for instance, when the former stands at 44° C. the second often reads less than 20°. On the other hand, in August and September, under the influence of the winds blowing from the Atlantic Ocean, the air becomes very damp and the heat grows stifling.

In spite of its excessive heat, the climate of Borkou is comparatively healthy; very relaxing during the hot and damp season, it is extremely

pleasant in the months corresponding to our autumn and winter. During my stay, lasting from 1913 to 1917, none of my European fellow-workers had any serious illness, and my black troops, though kept hard at work in the shape of arduous reconnoitring and escort duty, and with barely enough to eat, showed a percentage of sickness and deaths below the average of the other garrisons throughout the Chad Territory.

*Population and Commerce.*—The population of Borkou consists of nomads, the Tedas and the Nakazzas—the great nobles of the desert—and of a sedentary tribe, the Dozzas, who are only half noble, for want of the few camels whose possession would enable them to take a share in the profitable plundering raids in the desert. There is also a third category of inhabitants, the Kamajas, half serfs, half slaves, whose duty it is to attend to the gardens and the plantations of palms, and who are profoundly despised by the other two categories. The total population of Borkou would not appear to exceed some ten thousand souls, distributed among a score of more or less flourishing palm plantations.

The commercial activity of the oases of Borkou is far from negligible; they export towards the south salt, soda, and dates, and receive in exchange cereals, butter, cattle, and smoke-dried meat. Caravans of two hundred camels may often be seen coming to load up with salt at the Arouelli salt-pits near Ounianga; and Arab caravans pass by on the way from Cyrenaica, by Koufra and Sarra wells, importing to Wadai stuffs, sugar, coffee, tea, mercery, and (in time past) arms and ammunition; and exporting principally millet, butter, smoked meat, hides raw or tanned, ostrich feathers, elephants' tusks, and so forth. The slave-trade, formerly carried on through Borkou between Wadai and Cyrenaica on a great scale, has almost entirely ceased since we took possession of the country.

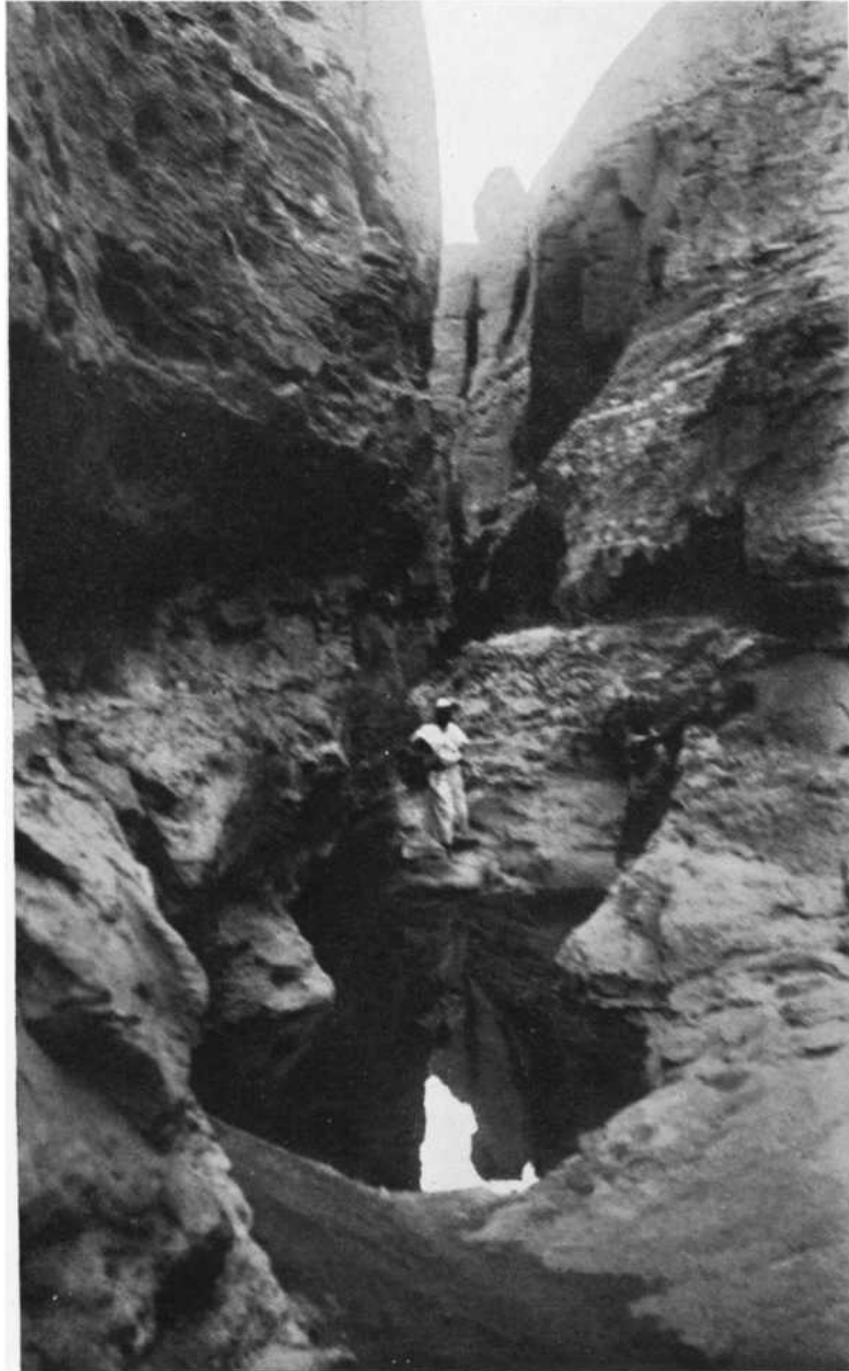
## 5. Exploration of the Western Borders of the Libyan Desert: Ounianga-Erdi

After drawing up the map of the western part of Borkou, subsequent to my reconnaissance in March and April of the various oases that succeed one another between Faya and Ain Galakka on the south and Gouro on the north, I devoted the last quarter of 1914 to an exploration of the unknown regions situated further east. Over and above their geographical interest, the said regions were of great military importance. My object was, in fact, to ascertain whether a counter-attack by the Senoussists, starting from Koufra and crossing the Libyan desert, could easily hope to escape the vigilance of our camel-corps patrols and fall on the remoter borders of Borkou and Ennedi.

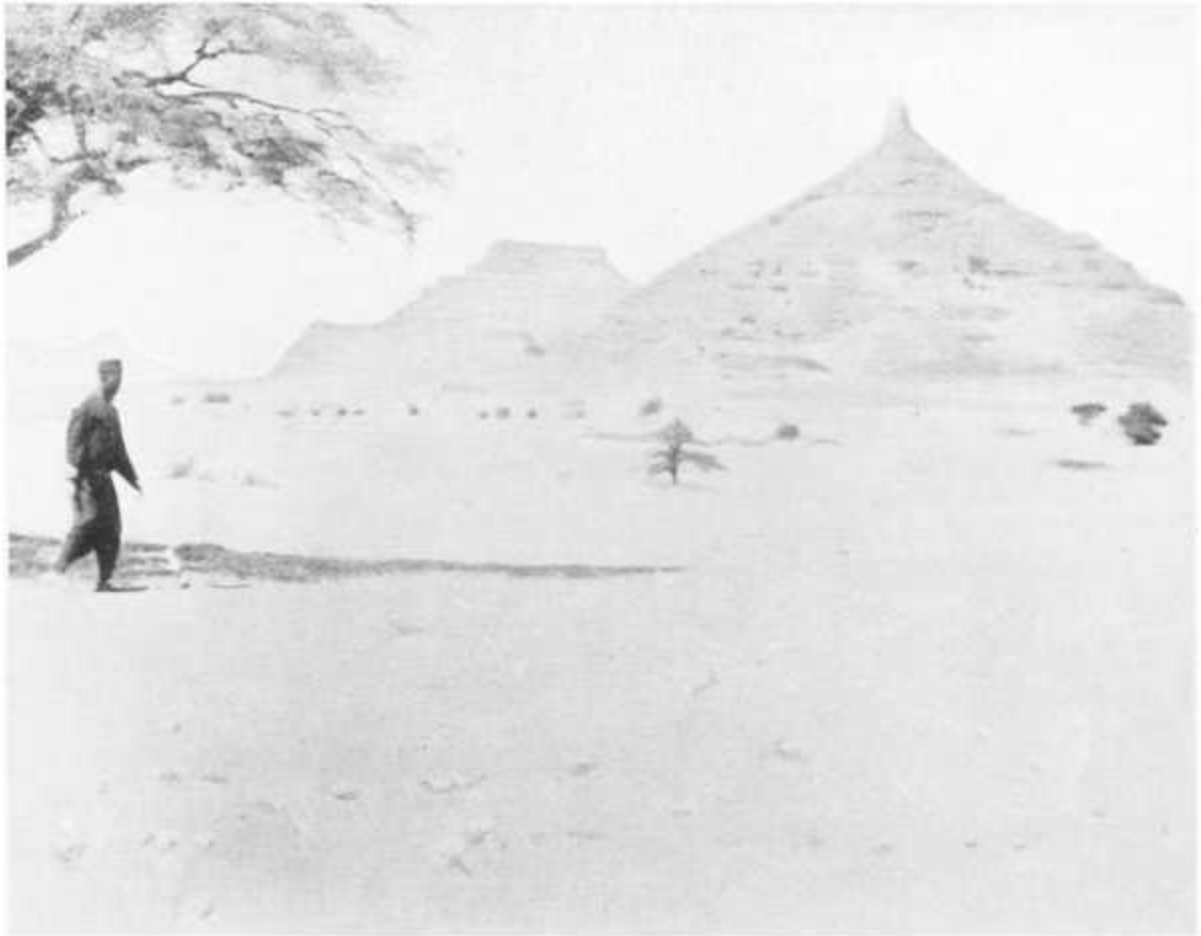
*From Faya to Ounianga.*—With this intention I left the oasis of Faya on 1 October 1914, at the head of a small escort, taking with me only some thirty lean camels tired and mangy, only capable of short stages and of carrying light loads. The result was that I spent nine days in covering the 117 miles between Faya and Ounianga, a journey that offers no difficulties and is usually completed in five or six stages. The points at which water may be found are frequent—at least one every 20 miles—and permanent; but grazing-grounds were almost non-existent at that time in consequence of the eleven years' drought the country had just suffered from. The rain that had fallen in August had, it is true, made a few green blades spring here and there, and they were eagerly snapped up by our camels as they passed; but they were still so scattered among the broken rocks that they rather emphasized than diminished the desolate barrenness of these dreary solitudes. From place to place, round a water-hole, one found a few wretched acacias, bushes of *rtem* or tufts of *akrech*. By chance one would come across what had once been a field of dried-up *hâd* whose thorny branches were grey with dust; but in a general way the landscape was disappointingly bare, and I wondered anxiously how long my camels would hold out on this starvation diet.

The route passed alternately through hamadas of sandstone, the blackened rocks of which emerged from irregular dunes, and through sandy plains into which one sank, raising thick clouds of dust finer than ashes. We did not meet a living soul on the way, except a detachment going back to Faya, and a little caravan consisting of two delegates of the Grand Senoussi coming from Cyrenaica on their way to Fort Lamby as an embassy to the commander of the territory. I spent an afternoon with them near the wells of Eddeki, and so had the pleasure of offering them tea. The chief delegate, Si Mahmoud Sheikh, was a Khoan of fairly high rank in the Senoussist confraternity. His appearance was that of a good Mussulman “brother” by no means indifferent to the good things of this world; fifty years old, and of a fine corpulence, he had a fair but sunburnt complexion, grey hair, a black beard, a round face, thin lips, small eyes, and a sensual nose. He was dressed all in white, walked with gravity, and spoke little. His attitude, free from arrogance, was not without a touch of awkwardness, and his reserve concealed but ill his uneasiness about the fate that might await him during his long journey among the infidels.

His companion, Abdallah Ghariani, was younger and of a very modest rank among the Khoans. He had a jovial, bustling manner, and talked volubly, but his eyes were sly and shifty. While we drank tea flavoured with mint, he boasted of the pacific intentions of Ahmed Sherif, insisted on the desire of the Confraternity to maintain active commercial relations between Cyrenaica and the Wadai, and on the necessity for suppressing the Toubou brigandage that hindered the march of the caravans. In conclusion, he declared that he had eaten no meat for a long time and begged me to make him a present of a small quantity of smoke-dried meat—a precious commodity in the desert, where the resources of hunting do not exist.



NATURAL CISTERN, ERDI



THE PEAK OF DIMI (600 m.), ERDI



THE PEAKS OF DOURDOURO (1000. m.), ERDI

*Ounianga*.—I reached the valley of Ounianga on October 9 in the morning, and was not a little astonished at failing to see the palm plantation till the moment of entering it; for, unlike those of Borkou, which can be seen from a distance, the oasis of Ounianga is hidden in a rocky excavation some 30 yards in depth and 4 or 5 miles long by 1 or 2 wide. The landscape thus formed is incomparably picturesque: a great sheet of calm water with blue shadows, edged with rosy-tinted beaches of sand, and fringed with green palm-trees stretched within a circle of bare wind-carved sandstone whose sombre hues cast here and there, under the blazing sun, warm shadows glowing with red or gold.

But it must be recognized that in spite of its beauty the palm plantation of Ounianga is but wretchedness, gloom, and disappointment. The inhabitants, known as Ounias, are few—some hundreds at most. On the other hand, millions of flies fiercely exercise their buzzing activity for

fourteen hours a day on man and beast. The soil is unfruitful, and produces hardly anything but dates. The foodstuffs necessary to life—cereals, butter, smoke-dried meat—are brought at great cost by caravans coming from Abéché to seek the supplies of salt from Arouelli needed by the inhabitants of Wadai. Even the camels cannot live in the neighbourhood for want of enough pasture, and from this cause our little garrison had the utmost difficulty not only in getting supplies, but in fulfilling the mission of watching the approaches of the frontier, and especially the great road from Koufra that emerges from the Libyan desert in the region of Tekro Arouelli.

It occupied at the north end of the lake a little rectangular fort, solidly built, but surrounded at a short distance by rocks that blocked the view and overlooked it to the north and east. It had not been possible to find a more favourable site, offering at the same time extensive views and an easily accessible water-supply.

I devoted two days to different tasks (inspections of the garrison, interviews with the Ounia chiefs and with two Khoans, former governors of the country in the time of the Senoussist domination, and so forth), and set out again on October 11 to visit the last water-points before entering the Libyan desert.

The Libyan desert is still almost completely unknown, no European traveller having been able as yet to cross it from side to side, whether from north to south or from east to west. In 1870 Gerhardt Rohlfs visited the northern part, as far as the oases of Koufra; a quarter of a century later British officers penetrated the south-eastern region as far as Bir Natrun, about 200 miles west of the Nile. On our part, we have been able to explore the south-western district and to obtain in respect of the central part fresh information, which it will not be easy to verify and extend until the French, British, and Italian governments combine in organizing for that purpose a geographical expedition, which would be of considerable scientific and even political interest.

I first took the direction of the salt-pits of Arouelli, situated 28 miles to the northwards, where I met a caravan that had just loaded up with 30 tons of salt for the Wadai markets. The salt-bed lies at the bottom of an absolutely bare sandy depression, covering some 25 acres. The bed of salt,

which is only about 6 or 8 inches thick, is on the surface, and more or less mixed with sand. The water-bearing stratum lies at a depth of 5 or 6 feet, and the water is naturally very salt. The water, rising to the surface by capillarity, evaporates, forming the salt crust that the caravans carry away in pieces, and which the natives of the Wadai and the countries bordering on it consume without further preparation. If one may trust the information supplied by the Ounias, the salt crust forms again about three months after being taken away, so that the output of the Arouelli pits would amount to nearly 100,000 cubic metres of salt annually, an output sufficient to satisfy the culinary needs of more than ten million people, and worth on the spot, as prices were before the war, some fifteen million francs.

From Arouelli I went eastwards to fix the position of the well of Tekro, where there is also a deposit of salt which is not worked, the admixture of sand being too great. The well of Tekro is particularly important, because it is situated at the extremity of the great caravan route joining the Mediterranean to the Soudan by the oases of Koufra and the well of Sarra. The water is abundant and fairly fresh, but the vegetation is reduced to a hundred clumps of siwak and a few tufts of grass of no value for the feeding of camels.

*The Route towards Koufra.*—Between Tekro and Koufra the distance to be covered is about 350 miles, about half of which had just been reconnoitred by Lieutenant Fouché, commanding the garrison of Ounianga. Marching in a general direction north-north-east he had first crossed a rocky zone of slight elevation, spending four hours in doing so; then for two days he traversed an immense sandy plain, bare of all vegetation, with here and there stretches of rock surface level with the ground; broken lines of rocky heights were visible in the distance to east and west. These heights went to join the plateau of Jef-Jef, in the direction of which he marched for twelve hours during the third day. On the fourth, he found himself in a vast plain from which the Djebel Habid, 50 miles away to the east, can be seen during the first few hours. The fifth day ranges of moving sand-dunes that served as landmarks for the guides were observed to the north-west, and at last, at nightfall on the sixth day, he reached the well of Sarra, lying in a hollow running from south-west to north-east and 30 metres deep.

The site of the well was chosen by the revered Sidi el Mahdi about 1898, and the works began almost at once. The boring, all done with picks and crowbars, was effected in hard reddish sandstone, by gangs of six workmen, relieved every month, and supplied with food and water by an endless succession of camel-convoys. At the end of eighteen or twenty months of uninterrupted work the water was at length found, clear, fresh, and abundant, at a depth of 80 yards, and since then the crossing of the Libyan desert has become relatively easy, the longest stretch without water being reduced to about 180 miles, whereas it was formerly almost 300. From the well of Sarra to Koufra the distance to be covered is only about 160 miles and offers no further difficulties, thanks to the intermediate well of Bechra.

What makes the journey from Ounianga to Koufra particularly troublesome is the total absence of pasturage for 500 miles, a state of things that results in the loss of many camels on every journey. The only good pasturage in the whole region is said to be found 80 or 100 miles to the east of the Sarra well, in the Djebel El Aouinat, an unexplored mountain mass of an extent not exceeding 1500 to 2000 square miles, as I am informed, and whose altitude may be roughly put at from 4000 to 5000 feet. It goes without saying that I only give these figures as a mere indication, and as subject to caution in every respect.

The break in continuity between the surveys of Rohlfs from the Mediterranean to Koufra and ours from the Wadai to the well of Sarra is consequently reduced to about 180 miles; but this gap does not seem likely to be bridged before Italy proceeds to an effective occupation of the oasis of Koufra, which falls within her sphere of influence.

Having ascertained the site, depth, and value of the Sarra wells, Lieutenant Fouché, in accordance with his instructions, set himself to march back to Ounianga, but the return journey was particularly dramatic. For from the very first day his guide led him directly south, instead of marching south-south-west. One is justified in supposing that he meant to lead astray in the desert the detachment whose camels were so exhausted that everybody went on foot, and whose store of water was limited to a little less than a gallon a day per man. Astonished at this unaccustomed deviation, the

lieutenant drew the guide's attention to it, but the latter answered: "Do not be uneasy, we are on the right road." But when he judged that the column was far enough from the tracks left by the outward journey, he replied to a fresh observation made by the lieutenant: "You are probably right, for I no longer see my usual landmarks; but if you would lend me a camel and a skin of water, I would go and find our tracks of the other day, and as soon as I had found them I would come back to look for you." The lieutenant thought it wiser to turn guide himself, and, compass in hand, he put himself at the head of the caravan, with what anxiety may be guessed! An error of direction of a few degrees—quite a usual thing in marching by the compass with no natural landmarks—might work out at a matter of 15 miles in a distance of 180, that being the distance to Tekro. And the well had to be found, in the immensity of the desert, before the detachment's scanty water-supply gave out! The black soldiers' thirst was aggravated by the crushing heat; reduced to a daily ration of a little less than 4 quarts of water, they no longer ate any solid food. The camels, grown weak, slackened their pace. The men, uneasy at not coming across their traces of the outward journey, thought themselves hopelessly lost. Their feet, swollen with weariness and made painful by the burning sands, seemed incapable of carrying them to the end of that interminable plain, torrid and unchanging, where the air vibrated as it vibrates above an overheated stove, creating all along the route deceptive mirages, ceaselessly dissolving and reappearing. After a while some of them lost heart and wanted to stop, preferring to wait for death where they were rather than go on with an aimless march. The lieutenant tried to cheer them up by singing the praises of his compass, and promising them that on the morning of the seventh day the three familiar rocks near the well of Tekro should appear before them on the horizon. Incredulous, but respectful, they betook themselves again to their journey, advancing automatically behind the camels as exhausted as themselves, and by some miracle, on the promised day and at the promised hour, they saw faintly outlined against the far horizon the rocks of their salvation! A few hours later, bivouacked round the well of Tekro, the brave fellows who had just covered 350 miles on foot in fourteen days in conditions of the utmost hardship, had forgotten their weariness and were contemplating with respect, on the lieutenant's table, the "good little iron" that had saved them from the most horrible death.

As for the guide, he was left unmolested, his criminal intention not being susceptible of absolute proof. It was the wisest course to take, for by punishing him without proofs, all we should have gained would have been to terrify men whom we might need later on! In the desert, the best guides may have their weak moments!

*From Tekro to Ounianga.*—From Tekro I came back to Ounianga, and continuing eastwards by the lakes of Little Ounianga and N'Tegdey I reached the salt-pits of Dimi, after crossing a chain of little sand-dunes about 50 feet high, stretching from north-east to south-west, and extending from 5 to 6 miles in breadth. This salt-pit lies in a sort of huge circle of rock, in the middle of which rises an isolated conical peak 500 or 600 feet high. It seems to me more extensive than that of Arouelli, but the salt from it does not seem to be so much in demand, on account of the very large proportion of sand it contains. The result is that it is hardly used by any one except the natives of Ennedi, who have only three days' journey to go in order to get a supply of it. The grazing, though by no means abundant, was less scanty than in the regions I had just come through, and my skeleton-like camels could eat their fill, for the first time in a whole month.

From the top of the rocks of Dimi my Ounia guide, Sougou, pointed out to me in the east the almost horizontal lines of cliffs forming the most westerly point of the mysterious plateaux of Erdi. The word "Erdi" means in the language of the Toubous "expedition, razzia," and would appear to have been applied to that region from time immemorial because it served as a meeting-place for the bands of raiders who put the caravans to ransom and pushed their raids as far as northern Dar Four and Kordofan, and sometimes even to the valley of the Nile in its middle reaches. According to the guide, rocky tablelands were to be found there, of an altitude comparable with that of Ennedi; the rains were less rare than in Borkou, the grazing-grounds for camels abundant, and the points where water could be found were hidden away in gorges difficult of access, little known, and hard to find the way to. For his own part, he hardly knew any except those of Erdi-Dji and Erdi-Ma, separated by a distance of 70 or 80 miles.

I hesitated some time before continuing my journey towards this region, whose very name was unknown till now; my water-barrels only gave me a

reserve of some thirty gallons, and my men's skin bottles were so corroded by the salts of sodium they had transported that they were empty after twenty-four or thirty-six hours' march. My camels, thin, worn out, and more and more mangy, could not do more than 20 miles a day, and I only had at my disposal ten days' supplies for my detachment, so that any error on my guide's part might put me into a critical position.

*Erdi.*—In spite of everything I resolved to make the attempt, trusting in fortune to ensure its success. In two marches we succeeded in reaching the foot of the cliffs of Erdi-Dji, 750 feet high and about 2000 feet above the sea. We found there good grazing for the camels, and from that day onward we had abundant fodder at each successive stage, so that I was delivered from the dread of seeing my indispensable beasts of burden waste away from inanition. The water was no less abundant, and was found in natural cisterns hollowed out by waterfalls in the beds of dried-up torrents that came down from the plateau. Some of these cisterns contained nothing but sand; but it was enough to bore a hole 1 or 2 feet deep in the sand to obtain a sufficient store of water.

From the top of the cliffs all that could be seen was an immense plateau, slightly undulating, and rising gradually towards the north-east. Beyond the line of the horizon some dozen miles away, there rose, as our guide told me, other cliffs; but all I could do was to take note of that information without being able to verify it.

Continuing our route eastwards along the foot of the cliffs, we reached five days later the region of Erdi-Ma, decidedly higher than that of Erdi-Dji: the highest altitude I had the opportunity of measuring exceeded 3000 feet. Our bivouac was installed at the entrance of the gorges of Dourdouro, where very picturesque natural cisterns are to be found containing abundant quantities of water withdrawn by the positions of the enclosing rocks from the drying action of sun and wind. During the whole of the way thither we did not see a living soul, any more than in the neighbourhood of Dourdouro.

My guide never having gone beyond that point, it was impossible to push my investigations further. Besides, I had now only four days' supplies left, a fact which obliged me to change my direction and make for Wad Mourdi, on the northern border of Ennedi, where I was to receive fresh

supplies. I had eventually to be satisfied with determining the position of this point and measuring a few heights while we were renewing our store of water before starting again after a day's rest.

This expedition, though limited to the south-western border of the massif of Erdi, revealed some interesting facts about the configuration of the country towards the 18th degree of latitude north and the 23rd degree of longitude east of Greenwich; the altitudes increased from west to east, and it seemed likely that the massif of Erdi was connected in one direction with the mountains of Tibesti by the plateau of Jef-Jef, and in another with the still unknown massif of El Aouinat, situated approximately between the 22nd and 23rd degrees of latitude north and the 24th and 25th degrees of longitude east.

Later information gave me a few further indications about western Erdi, where two water-points were found; one Bini-Erdi, about 80 miles north-east of Dourdouro, and the other, Erdi-Fouchini, some 60 miles north of Dourdouro, at the foot of a line of tall cliffs. The deduction may be allowed, for the time being, that the central tableland of Erdi offers altitudes presumably superior to 4000 feet, and that it slopes gently down on the east to the great sandy plain, without vegetation or water, across which passes the route from El Aouinat to Merga, a route that establishes direct but very difficult communication between Koufra and Dar Four, to the east of the 24th degree of longitude.

*Between Erdi and Ennedi.*—In leaving Dourdouro to march southwards I was going into the unknown. I could, no doubt, see in front of me, 40 miles away, the crests of northern Ennedi, at the foot of which I was to find the water-points of Aga and Diona; but to seek the said points without guide in the chaos of rocks was a risky undertaking, and might have been held unreasonable if the way our supplies were running short had not obliged me to go forward.

A vast depression, stretching from south-south-west to north-north-east and of an average breadth of some 30 miles, separated Erdi from Ennedi; it was the depression I heard spoken of earlier as a prolongation of that of the Bahr El Ghazal, through which Lake Chad once poured its waters into the lakes of Toro and Djourab, and consequently that by which the basins of the

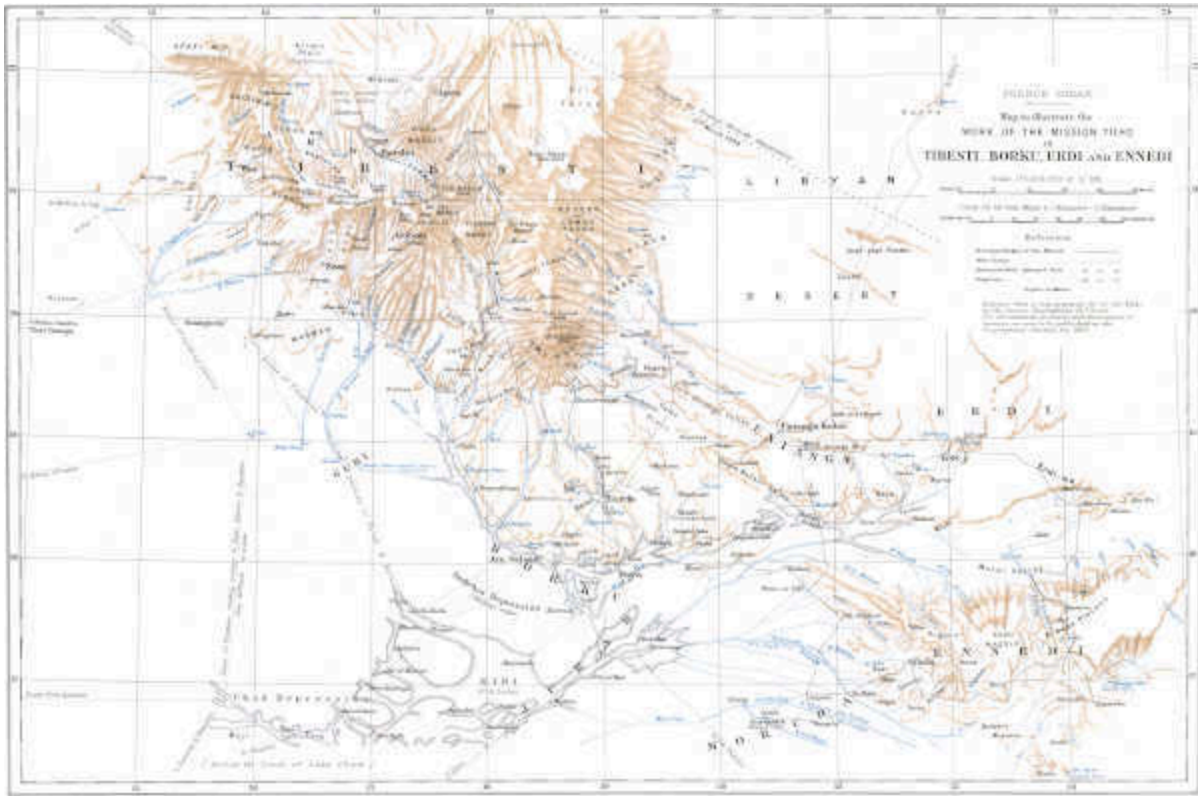
Chad and the Nile might in ancient times have entered into communication. That being so, I took the utmost care in examining the region and determining the altitudes. The lowest point was found about 30 kilometres from Dourdouro. Its altitude was 1750 feet, or 1000 feet higher than that of Bokalia at the north-eastern extremity of the Djourab. The slope was therefore from north-east to south-west, as was confirmed by the shape of the ground and the general direction of the valleys running into that depression, and I was able to conclude that if an ancient river once flowed in the bottom of that broad valley, which is hardly likely, it ran, not towards the Nile, but towards the lowlands of the Chad. By this evidence, one of the most important items of my geographical programme was fully elucidated: the basin of Lake Chad constitutes in the centre of Africa a closed basin which has never been connected with the basin of the Nile. The lake zone, now dried up, consisting of Kanem, the lowlands of Lake Chad, and Borkou, was once the outlet for the affluents of Lake Chad and for many great rivers coming down from the mountain mass of Ennedi, Erdi, and Tibesti. Its outline at successive periods—an outline in all probability very irregular—might be indicated by the hypsometric curves 270—260—250 metres, adopting for the Lake Chad of to-day the altitude of 240 metres. Its extent at that period must have been comparable with that of the Caspian Sea at the present day, and its greatest depth some hundred metres.

In the evening of the second day's march, when we were drawing near the foothills of Ennedi, we had not yet found any well, and our tiny store of water was used up. But spying in the west a notable gap in the line of hills, I thought we should be likely to find a water-point there, and profited by the coolness of the night to try to reach it. At dawn we came out on a fine river, dried up, where we got a little water by digging holes in the sand. By good luck our guide, Sougou, recognized that we had reached Oued Mourdi, where he had come by another route some six months earlier; thanks to which discovery, after a little search we were able to bivouac beside the well of Diona.

If I had had time and means, it would have been extremely interesting to explore up to its starting-point the great depression I had just crossed, a depression which perhaps comes down from the region of Merga in the heart of the Libyan Desert, where the natives agree in declaring that there

exists a little lake surrounded by a palm plantation. The probable position of Merga is between the 25th and 26th degrees of longitude east and 18th and 19th degrees of latitude north. This oasis is situated on the direct route from Ennedi to Dongola, about 200 miles from the last water-point of Ennedi (Gourgouro).

FRENCH SUDAN  
 Map to illustrate the  
 WORK OF THE MISSION TILHO  
 in  
 TIBESTI, BORKU, ERDI AND ENNEDI  
 THE GEOGRAPHICAL JOURNAL, AUG 1920.



*Modified Polyconic (1/M. International Map) Projection.* Published by the Royal Geographical Society. TIBESTI Tilho

[\*\(Large size\)\*](#)

## 6. Exploration of Ennedi.

Having reached the well of Diona on 11 November 1914 in the morning, I was joined next day by the camel-corps section of Borkou and Ennedi, which brought me fresh supplies and were charged with the mission of getting into touch with the nomads of eastern and central Ennedi, who refused to acknowledge our authority and committed acts of brigandage on our lines of communication. A few patrols in the neighbourhood having made it clear that the rebels had decamped before us and taken refuge on the high plateaux, the camel corps under the command of Captain Châteauvieux climbed the heights of Erdébé, where they began an active pursuit of the rebels. At the same time I reconnoitred the water-point of Aga, 30 miles further east on the route from Erdi to Dar Four, a route followed at that period by a certain number of Senoussist emissaries on their way to exhort the Sultan Ali-Dinar to join in the Holy War! For it will be remembered that Turkey had just at that date entered into the war against us, and that the plan of the German general staff included a vast Musulman rising destined to drive the French and British out of their African possessions.

*Eastern Ennedi.*—Finding no traces of the rebels at Aga, I rejoined the camel corps in their occupation of the cisterns of Keïta on the plateau of Erdébé, and until the end of November our reconnoitring columns explored the labyrinth of gorges and rocky valleys over which the refractory natives had scattered, without offering serious resistance anywhere. The cold was beginning to be rather unpleasant, especially when the north-east wind blew, but the thermometer did not fall as low as zero. The water-points were extremely numerous, a fact which favoured the break-up into small fractions of the rebel bands, whose chief anxiety appeared to be the getting of their herds of camels and oxen and their flocks of goats into a safe place. They did not seem to worry much about their women and children, and let us capture them with the serenest unconcern, being well aware that we should do them no harm, and that their sustenance would be assured for the time being by our black troops, always glad to leave the preparation of the

daily cousscouss to the other sex. To conclude this series of operations we had to fix the limits of eastern Ennedi. An expedition was sent to Bao, 60 miles southwards, the last water-point in the region, and thence to Kapterko in the south-east, where a few rebels were captured. Another expedition fixed the position of the well of Koïnaména some 50 miles east, and went a stage further, to the beginning of the great plain without water or vegetation that stretches out of sight to the eastward.

The general physiognomy of the country was that of a rocky tableland intersected by a great number of valleys, more or less deep, and gorges, separated by many little jagged chains of sandstone running in all directions, and varying in height between about 200 and 500 feet. All those depressions are covered with grass and shrubs, affording excellent pasturage for the hillman's flocks. Of plants useful for human food we found gramineæ such as the *Kreb* and *Anselik*; what is more, the soil of the valleys was literally covered in places with water-melons and colocynths. Though I found no traces of tillage anywhere, I even had the surprise of noticing from time to time hardy stalks of the wild cotton plant, some reaching 6 feet in height.

Almost every year at the end of the rainy season temporary rivers flow through these depressions, some of them turning northwards (and consequently tributaries of the Chad basin), the others southwards, where they once used to feed some great tributary of the Nile basin. Numerous pools formed during the rains hold out for a longer or shorter time in the flats of the more considerable of these valleys, while in the narrower parts the water is stored in natural reservoirs, more or less hard to get at, hollowed in the sandstone by the falling waters as each torrent makes its way down from one ledge to the next.

The greatest altitude I noticed in the course of my surveys on the plateaux of Erdébé was found in the water-parting between the slope towards the Chad and the slope towards the Nile: it was of 3600 feet. The highest summits in the neighbourhood rising only from 250 to 400 feet above the general level of the country, it may be estimated that the chief altitudes of that region vary between 4000 and 4200 feet. Twenty miles east of Koïnaména, in the transition zone between the mountains and the plains,

the altitudes of the bottom of the valley was still superior to 3000 feet. It is possible, moreover, that 40 miles away to the north-east certain summits of the water-parting rise to 5000 feet.

The natives who live a nomadic life on the plateaux of Erdébé amount in number to several hundred families. Their settlement, meagre in the extreme, usually consists of a few pieces of matting stretched on stakes in a corner of a ravine, round a thorn enclosure in which their flock of sheep and goats is shut up; at the slightest alarm men and beasts stampede among the rocks. If I had to seek in the animal kingdom a term of comparison for these tribes, I think I should choose their fellow-denizen the jackal: they possess its cunning, its audacity, its cowardice, its mischievousness, its endurance, its speed, and its predatory instincts.

The only other wild animals we saw were gazelles, antelopes, and ostriches; it is reported that as long as the above-mentioned pools remain, boars, panthers, and lions may be found, but we had no opportunity of testing the truth of this assertion.

On December 9, in the afternoon, having made preparations for our departure next morning, we set free our prisoners, imposing no conditions beyond that of telling their fellows our desire to see peace and quiet reign throughout the country. "Let the nomads devote themselves to the raising of their flocks and to trading in salt and millet," I said; "let them give up raiding the peaceful tribes of the Sudan and the Nile, and the caravans that cross the desert, and I will leave them at liberty in their mountains." Whereupon an old woman answered me, "We will carry your words faithfully to our husbands and sons, and we will bid them come and submit to your authority; we are all weary of our perpetual insecurity; we desire peace and justice. You have treated us well, you have given us millet and meat; we have eaten all we wanted to eat, and now we know that you are strong and generous. Allah reward you!"

Alas! my reward was that for two years longer these inveterate brigands did not cease raiding in every direction, and that the camel corps had a particularly difficult task in guarding convoys and putting down pillaging.

*Western Ennedi.*—It only remained to me to cross the central part of Ennedi in order to have a clear outline of the general physiognomy of the country, thanks to the aid of surveys previously executed on its western borders by several officers who had taken part in military operations in Western Ennedi under the orders of Major Hilaire and Major Colonna de Léca. With this end in view, I marched in the direction of the military post of Fada by Boro and Archeï.

For a week our route lay through a maze of sandstone rocks where no track existed, and through which our guides zigzagged from crest to crest with remarkable sureness. Sometimes we made a long *détour* to cross a wadi near its source; sometimes we marched straight for the obstacle, dropping down steep ledges that inspired little confidence in our animals, or crossing difficult ridges that the camels could only climb after being unloaded. Everywhere were narrow gorges and jagged crests, with here and there a few leagues of easy going in the neighbourhood of the temporary pools that usually marked the convergence of certain important ravines.

In this uneven ground with its narrow horizons one pasture-ground succeeded another, but we saw no trace of inhabitants. And yet water was not wanting, whether in natural cisterns or in great pools like that of Kossom Yasko. We skirted on the south the tableland of Basso, higher, according to our guides, and harder to climb than that of Erdébé, but, so far as I could judge at a guess, its height is not likely to be as much as 5000 feet.

We took a day's rest in the excellent pastures of Boro before leaving the central plateau of Ennedi to drop down to the next level, 400 or 500 feet below. Then our way lay along a fine river of white sand, between banks 60 or 80 yards high, where the traces of the last flow of water could be seen 6 or 7 feet up the bank. The coming of the floods is so sudden, and the banks so steep and smooth, that it is dangerous to take that road in the rainy season. No winter passes without some heedless wayfarers being surprised and carried away by the rushing torrent that comes sweeping down the valley with the speed of a galloping horse.

After this splendid sand-road came a stretch of rocky going, followed by a zone of waterfalls we had to get round by a march on the plateau. The

lower we got the more picturesque the landscape became; the cliffs, gaining in height what we lost in altitude, grew more and more imposing, the crests more jagged, the ridges more often broken by gaps. Isolated peaks appeared here and there, whose pure outlines and bold summits put climbing out of the question. On all sides there rose in the distance rocks, some broad, some slender, but all of the same height and grouped irregularly, so that sometimes, when very close together, they looked like groups of men.

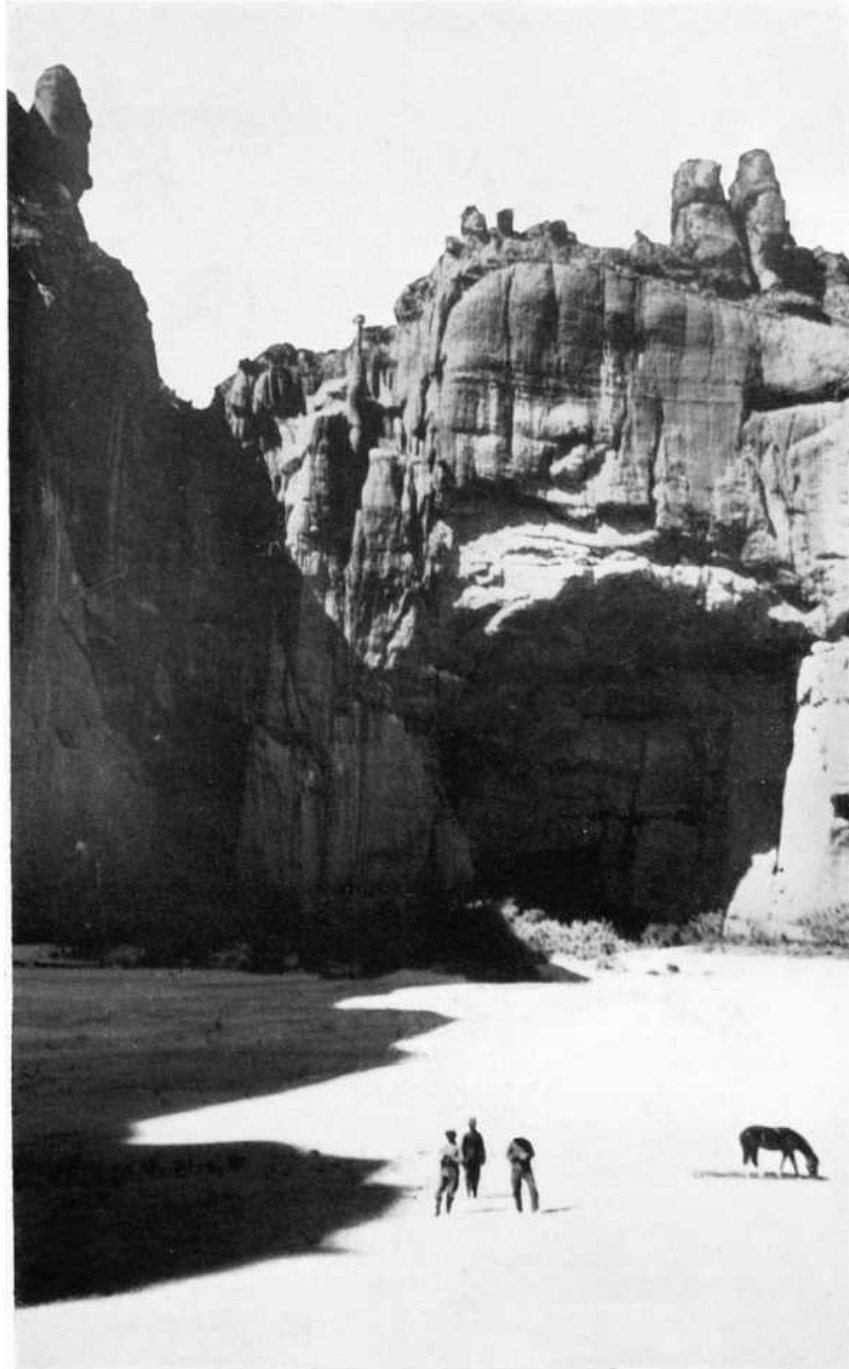
On the 17th of December we reached the foot of the last ledges, on the western borders of Ennedi, at the altitude of about 1800 feet—that is to say, about that of the depression separating Erdi from the plateaux of Erdebe—and pitched our tents in the valley of Archeï, the most picturesque of the beautiful valleys of the Ennedi. The century-long erosion of wind and water, carving the great sandstone masses that line the valley, lavished throughout the landscape the most admirable effects of natural architecture. The approaches of the great grotto, above all, and of the sheet of water teeming with little fish, were a pure delight for the eyes: the sheer cliffs, fretted into colonnades crowned with turrets and belfries, were burnt to tones of faded ochre that made the blue of the sky seem deeper and more luminous still.



MOURDIA WOMEN AND CHILDREN, PLATEAU OF ERDÉBÉ (1000 m.), ENNEDI



THE FORT OF FADA, ENNEDI



CAVES OF ARCHEÏ, ENNEDI

From this exploration it became apparent that Ennedi is, roughly speaking, a triangle covering about 12,000 square miles (30,000 square kilometres). It consists of a succession of sandstone plateaux rising in tiers from the base level of 1600 feet to that of 4300 and possibly even 4800 or 5000 feet in the parts of the country which had to be left out of our

investigations (Basso and eastern Erdébé). It falls by steep slopes to the plains of the Libyan desert. The plateaux of Ennedi are ravined by many valleys, most of them very deep, whose waters only flow for a few days or weeks each year after the rains (August and September). These waters hurl themselves from ledge to ledge in waterfalls, hollowing out at the foot of each fall natural cisterns in the rock, where the water remains a longer or shorter time according as it is well or ill sheltered from the torrent beds. The roads usually follow the torrent beds, except when blocked by masses of crumbled rock, in which case a more or less awkward circuit has to be made. At the points where the main valleys converge great muddy ponds are usually formed, but they are shallow and short-lived. In all the valleys splendid grazing-land is found, where not only camels but also thousands of oxen could live if the problem of drinking-troughs did not present itself every year in the height of the dry season. For at that moment the natural cisterns that have still kept some store of water are grown few in number, and are nearly always very hard to get at. Most of the great temporary pools are dry, and subterranean water is no longer found except in the great wadis, where the wells (that have to be dug out afresh every year) go as deep as 20 or 25 yards.

The inhabitants of Ennedi, nomads or semi-nomads, are very poor; the chief tribes are the Bideyats (or Annas), the Gaedas, and the Mourdias, which all together represent hardly more than 2000 souls. But they are by tradition so addicted to brigandage and so untamable that as large a troop of police is needed to keep them in hand as for a population of 40,000 in the settled regions.

Ennedi has no vegetable food resources; there are neither palm plantations, nor native gardens, nor millet fields. And yet the soil is more fertile than in Borkou and the periods of drought shorter. The chief agricultural interest of the region lies in its excellent pasture, where the camels find abundant provender of very good quality.

*In Mortcha.*—From Archei I went to the post of Fada, 40 miles or so to the north-west, for a few days' rest, after which I undertook a new series of reconnaissances westwards, for the purpose of exploring the still imperfectly known desert regions of northern Mortcha, too often visited by

the raids of the refractory tribes. I was thus enabled during the early days of January 1915 to trace the course of the temporary rivers that receive the waters from the western slopes of Ennedi. For a few days every year these rivers roll down a volume of water sufficient to stop the march of caravans and convoys for a longer or shorter time, and continue their course for 200 or 300 kilometres before each of them reaches the pool in which it ends. As they have not force enough to go further, all one finds beyond the terminal pool is a valley-way more or less clearly marked, and blocked with sand from place to place, but still visible for fairly long distances. It has been concluded that they formerly ran into the ancient lake of Djourab, the level of which is from 200 to 300 yards lower. The most interesting of these rivers from the geographical point of view is the wadi Soala, which in the central and lower parts of its course separates the granitic zone of Mortcha from the sandstone of Ennedi.

The whole region is one succession of good grazing-grounds for camels, but which can be made use of only a few months a year while there is water in the temporary pools. The one that lasts longest, that of Elléla, in which the wadi Oum-Hadjar comes to an end, is not entirely dry till April or May when the annual rains have been normal, in which case it makes direct communication possible between Borkou and Wadaï.

*Between Ennedi and Borkou.*—I next set out northwards from Ennedi in the direction of Madadi and Wadi-Doum, which had been adopted for the time being as their headquarters by some rebel bands from Tibesti, which attacked indifferently the caravans from Wadaï going to Arouelli for salt and our unescorted convoys of supplies circulating between the posts of Faya, Fada, and Ounianga. At the moment when I arrived in the neighbourhood they had just carried out successfully several of these surprise attacks, and were making off to their mountains to get their booty into a safe place. Unable to go after them, for my camels, exhausted by three months' reconnoitring and hard fare, could not challenge those of the rebels for speed, I decided to return without delay to Faya to organize reprisals.

On the way I passed through a low-lying zone of country once occupied by lakes and marshes of considerable extent and of about 1000 feet in

altitude, or 250 or 300 feet higher than the region of the ancient lakes of Borkou and Djourab, with which it is connected by a continuous valley, the bed of which, very clearly visible in places, is often buried in sand. This lake-zone seems to be the end of the great depression I had crossed two months earlier, between the massifs of Erdi and Ennedi. Except in the immediate neighbourhood of the springs of Madadi and around the permanent pool of the Wadi Doum (or Touhou) the soil is absolutely barren, consisting either of very pure siliceous sand or of soft friable earth, whitish in colour and as fine as flour, into which we sank to the ankles at every step, raising thick clouds of stifling dust. Towards the south stretched chains of shifting sand-dunes, separating that depression from the last foothills of Ennedi, while to the north extended endless rocky terraces, in which were hollowed here and there basins of 1 or 2 square miles, wells of water impregnated with soda.

*The Holy War.*—The Turco-Senoussist propaganda against the French and English was beginning to make its pernicious effects felt among the nomads of Borkou and Ennedi. The easy successes achieved by the rebels against caravans and convoys unprotected by escorts had just given them a great idea of their military power, and increased their numbers and audacity. The withdrawal towards their base of the Italian forces in Tripoli, and particularly the abandonment of Mourzouk, where a Senoussist governor had taken up his residence, had inflamed the minds of the Toubous, whose warlike ardour had never burnt so fiercely: it seemed to them likely that a backward movement of the French occupying Tibesti, Borkou, and Ennedi would speedily take place if their commissariat lines were seriously threatened in the direction of Lake Chad and Wadaï. Turkey's entrance into the war on the side of Germany against France and England had counterbalanced the successes won over the Germans in the Cameroons and deeply stirred the imaginations of these devout Mohammedans, who refused to recognize any other chief than the distant Sultan of Stamboul, Caliph of the Prophet and Commander of the Faithful. And one after another the Duzzas of Borkou, the Gouras of Gouro, the Arnas of Tibesti, and the Gaïdas of Ennedi fell from their allegiance.

Now, at that moment the requirements of the escort-service for our convoys of supplies were such that out of the hundred and sixty men of

each of my companies in Borkou and Ennedi, less than twenty rifles were sometimes left to guard the posts of Faya and Fada. It was hardly before the month of April 1915, when the food-transport was almost finished, that it became possible to remedy this dispersal of our forces and organize the punitive expeditions rendered indispensable by the incessant raids of the rebels. That task was an awkward one, for we were short of good camels and above all of good agents of information, while our elusive adversary was kept acquainted with our slightest movement by certain elements of the population theoretically faithful to us.

It would evidently have been too much for us to hope that we should speedily obtain the submission of the malcontents, given the very considerable extent of their space for movements of all kinds, and also their extreme mobility; but we could henceforth return blow for blow, chase them to their mountain lairs, and give them the impression that, after playing for some time the pleasant part of hunters, they were henceforth going to play the much less pleasant one of game.

One after another Captains Lauzanne and Châteaueux, Lieutenants Lafage and Calinon, at the head of mixed detachments of regular soldiers and Arab and Toubou auxiliaries, made their way into the wildest fastnesses of Eastern Tibesti, Borkou, and Ennedi. Captain Lauzanne, in particular, succeeded in tracking the Gourmas into the distant solitudes of Ouri, 200 miles north of Gouro, at the foot of the eastern spurs of the Tibesti, and after them their cousins the Koussadas into the very crater of Emi Koussi, till then regarded as impregnable. The fame of these two expeditions was noised abroad in the country to such an extent that by the end of the month of July the general situation of Borkou had greatly improved, and we could turn our thoughts to the consolidation of our prestige by an offensive action against the rebels of Miski, and by a junction of our troops with those of Zouar and Bardaï, the two military posts entrusted with the supervision and pacification of western and central Tibesti.

## 7. Exploration of Tibesti.

In the month of September 1916 I was authorized to proceed from Borkou to Tibesti for the purpose of getting in touch with the rebel tribes who intended to attack the caravans fitted out in Kanem and Wadaï for the carrying of supplies to the garrisons of Borkou and Ennedi. The garrison of Tibesti was to attempt, to the best of its ability, to co-operate with this action in such a way that the hostile bands, threatened at once on the south, the west, and the north, might either be induced to submit or else to disperse in the eastern part of the Tibestian massif, the part furthest away from the region to be traversed by our convoys of supplies.

The rebels were comparatively few in number—about 2000 combatants—and divided into clans living in different regions; but they were of extreme mobility, well armed, and abundantly supplied with ammunition. Their tactics, which were very skilful, consisted in avoiding on all occasions a fight in the open, in hiding in the labyrinth of their well-nigh inaccessible rocks to fire at short range on the enemy when he passed near enough, in decamping at top speed to hide again a little further on, and so draw little groups of adversaries in the direction of death-traps, where of course well-planned ambushes lay in wait for them.

The strength of the reconnoitring detachment was forty-four black soldiers, officered by four Europeans—one of them a doctor—and accompanied by some thirty auxiliaries (guides, goumiers,<sup>[1]</sup> camel drivers, and servants). It carried food for two months, and the barrels and skins required for three days' water. The train included about 120 camels.

The mountainous country to be crossed set an extremely awkward problem: many points where water would have to be found were often hard for the camels to reach. Pasture-grounds were rare and scanty. The tracks, inexistent or deceptive, would now stretch away across successive heaps of sharp-edged pebbles, and now twist and turn endlessly along winding torrent beds, deep sunk between sheer banks. To cross from one valley to the next one had to climb a succession of cliff ledges, rising tier on tier to

several hundred metres by the merest suggestion of paths winding along the sides of spurs formed by the rolling down of *débris* from above; when the slopes grew too steep, the baggage had to be carried up from one shelf to the next on men's heads. Our camels, used to the easy going of the great sandy plains, were discouraged by the asperities of the sharp-angled rocks, by the narrow ledges, the steep and slippery steps, the loose pebbles, the excessively sharp turns; and so only short distances could be covered in spite of long hours under way and intense fatigue.

It goes without saying that we had no sort of map of these unknown regions, and that we were utterly at the mercy of the guides whom by good or evil fortune the patrols put at our disposition. Accordingly, the choice of our routes was dictated to us at once by the necessity of reducing to a minimum the efforts and privations of our camels and by that of keeping within the limits familiar to our ordinary and occasional guides. It may be added that the latter showed the utmost unwillingness to lead us into regions where the unsubdued tribes habitually take refuge; for these tribes are in the habit of holding them responsible, on their own heads and those of the members of their families, for all the harm and losses incurred when fights arise with our detachments.

The general plan of this series of operations included, first of all, the reconnoitring of Emi Koussi, an extinct volcano 3400 metres high, followed by an inroad into the valley of Miski, the usual meeting-ground of the Tibestian freebooters threatening the roads to Kanem. The central position of the valley is strengthened by the natural shelter afforded by high mountains and almost impassable rocky foothills, through which lead only two defiles, both of them long and dangerous.

From Miski I meant to make a rapid plunge into the valley of Yebbi, in the heart of central Tibesti, firstly to try to get into connection with a detachment of the garrison of Bardai, and then to make an attempt to reach the plateaux of Goumeur. Lastly, I thought I might be able to get over on to the western slope of the massif, explore its chief valleys, and effect a junction with the Zouar camel corps before returning to Borkou. I succeeded in carrying out this programme in its main lines, except for the operation in the direction of Goumeur, which had to be replaced at the last

minute by a reconnaissance pushed as far as the post of Bardai. I was away, in all, for seventy-two days, or barely a fortnight in excess of my estimate.

*From the Plains of Borkou to the Foot of Emi Koussi.*—The name of Borkou is given by geographers to the group of low-lying stretches of country separating the mountain mass of Tibesti from that of Ennedi; it was confined at first to the depression, some 10 kilometres wide by 100 in length, that extends from east to west, from Faya to Ain Galakka.

This hollow was long filled by a lake, of which numerous and conclusive traces are still found: beds of lake shells, whole skeletons of fishes up to a yard and half long, calcareous crust covering long streaks of rock, platforms of white clay marking the line of flats where the last pools left by the waters of the former lake have held out longest before drying up, and so forth. This lake was fed by mighty watercourses, coming down from the mountains of Tibesti and Ennedi; it poured its overflow through the valley of the Jurab into the Kirri, the deepest, largest, and most recently dried up among the ancient lakes and lowlands of the Chad.

From Borkou to Emi Koussi there is a large choice of routes. The best, owing to the number of points at which water and pasturage may be found, is that which passes by way of Yarda to Yono. Hereabouts we leave behind the region of the oases characterized by numerous depressions in which water is found close to the soil in practically unlimited quantities, in wells less than a yard deep and in salt pools. From that point one enters the rocky zone where there is no more water underground, but only natural cisterns forming reservoirs with the water that streams down into them, and dries up a longer or shorter time after the passage of the accidental rains that filled them.

The general look of the country is fairly uniform. It is a vast sandstone plateau sloping from north to south, ravined with narrow gullies running in a general direction from north-east to south-west, and which are real rivers of sand in which the shifting dunes pile themselves up and overlap to the point of being impassable at times to laden beasts of burden. This direction, from north-east to south-west, being that of the prevailing wind in Borkou, the parallelism of these gullies and the general appearance of the landscape

give colour to the supposition that they were hollowed out of the sandstone by the erosive action of the dunes driven before the wind.

The rocky plateau is commanded at intervals by a few blackish peaks of low relief, among which the most noticeable are those of Kazzar, near Yarda, 75 metres above the surrounding country; Olochi, near Dourkou, 130 metres; Ehi Kourri, near Kouroudi, 350 metres in relief. From the height of these natural observatories nothing is to be seen, in whatever direction one turns, but vast dark-tinted expanses strewn with stones, where no sort of topographical order can be discerned. So confused and scattered are the rocky masses that the impression they leave is less that of a sequence of alternating plateaux and valleys than of a chaos of disconnected reefs rising above a sea of sand, amid breakers of billowy dunes. Much going and coming was needed before I could form an exact notion of the physiognomy of these regions, for the fact is that their valleys are more or less blocked, at longish intervals, by heaps of rock debris and sand, and so divided into a succession of elongated hollows communicating only by subterranean infiltration. In these hollows may be found, here and there, layers of shells that enable us to fix the period when they were still underwater at a comparatively recent and no doubt Quaternary epoch. From place to place there still exist permanent salt pools, of greater or less depth, and usually at the foot of the cliffs that shut in some of these valleys on the east. One supposes that the strong back draughts of the north-east wind have mainly concentrated their action on those points of the surface where the sandstone was softest; in the excavations thus produced the sheet of subterranean water has been able to make its appearance in the open air, and under the influence of a persistent evaporation, due to the extreme dryness of the air and the intensity of the solar heat, the salts in solution in the water have undergone a progressive concentration, sometimes to the point of floating on the surface of the pool with the appearance of translucent blocks of ice.

Having left Faya on September 4 we arrived on the 11th at the foot of Emi Koussi, 125 miles to the north, passing on our way by Korou Koranga, where we renewed our supply of water. The spot is one of the most picturesque I saw during this journey to Tibesti; it is a natural cistern

hollowed by the action of the falling waters in the deep and narrow bed of the wadi Elleboe, a torrential river that comes down from Emi Koussi. The way to it lies through a defile more than a mile long, so narrow that two men cannot walk abreast. The water lies at the bottom of a grotto, dark in spite of being open to the sky, and whose walls wind in and out in such a way that not only the drying desert winds cannot get to it, but that even the sun's rays only penetrate to it for a few minutes each day about noon, and only get down to the level of the water during May and July, when the sun reaches the local zenith. I had neither the time nor the means to measure the length and depth, the approach between precipitous walls being so difficult; but the supply of water is such that the cistern has never been dry so long as the guides can remember, however long may have been the drought during which the torrent has ceased to flow; the water stays clear, cool, and pleasant to the taste, without the slightest salty flavour.

The cistern of Derso, on the contrary, at the foot of Emi Koussi, near the pasturage of Yono, is broad, spacious, and subject to the drying action of sun and winds; a score of yards deep, it is easy to get at; but its greenish water, stagnant and thick with organic matter, has to be filtered before it can be drunk without disgust, and a period of twelve or fifteen months' drought is usually enough to dry it up altogether.

*Ascent of Emi Koussi.*—In all probability the rebels of the regions we had just come through had withdrawn towards their strongholds on the top of Emi Koussi. A light detachment was sent out to make sure that this was so, while the greater number of our camels were left to rest in the pasturage of Yono, where I had a little zeriba built for the storage of our baggage and provisions and the security of the men I left to guard them.

On the morning of September 13 we betook ourselves to the ascent of the mountain by a track strewn with boulders, the gradient being fairly easy for the first five hours' march, as far as the salt springs of Erra Shounga. From that point it stiffened, and grew very steep indeed between 6000 and 9000 feet. The last part of the ascent to the entrance of the pass that leads into the interior of the crater required the utmost effort on the part of our camels, unaccustomed as they were to the going in mountainous countries.

Sixteen or eighteen hours must be allowed to reach the summit of the ancient volcano, and one does well to spread them over two days if one does not want to leave any camels on the way. The first stage should get one to Fada, a little pasturage at the bottom of a ravine accessible to camels, and where the animals should be allowed to rest and feed. Afterwards a fairly long halt should be made at an altitude of about 6000 feet, to renew the supply of water at the natural cistern of Lantai-Kourou, for there is no hope of finding water in the interior of the crater; the operation is a long and toilsome one, for the track leading to the reservoir is inaccessible except to men. Along the whole way there is hardly any vegetation, such as there is being confined to deep ravines, almost always inaccessible, except at the pasturage of Fada, on account of the steepness of their sides. Towards the foot of the mountain only stunted plants are to be found, with tiny leaves often sharpened into thorns; while nearer the top the boughs are thicker, the bark tenderer, the sap more abundant, and the leaves longer and greener. No trees are to be found on Emi Koussi in the crater itself; on the other hand, the herbaceous vegetation is comparatively abundant, and marked especially by the "erendi," a yellow-flowered plant reminding one of the St. John's wort of our regions. We bivouacked, in a good position for observing all the approaches, in the midst of these bright-hued flowers, and I cannot tell you with what fascinated eyes we gazed on them, for none of us had seen their like for three long years.

The temperature was mild and cool like that of a fine spring in France; but in the clear sky there were no birds, and the sight of the scowling cliffs around us soon broke the charm under which our fancy would have gladly lingered.

We stayed only three days in the crater of Emi Koussi. The afternoon of the first day was devoted to the exploration of a pit, 300 yards deep and 2 miles in diameter, which was once the chimney of the volcano. A vast expanse of carbonate of soda covers the bottom, which one can reach only by a very steep path.

The second day was spent, firstly in exploring, both inside and out, the western slopes of the crater, where there is a natural cistern that enabled us to make a fresh provision of water, though the track leading to the reservoir

is very perilous for the camels; and afterwards in taking certain measurements, such as the height of the cliffs and the depth and extent of the central pit, called by the natives Era-Kohor, or Natron Hole.

The third day was given up to explorations in several directions, which allowed us to visit some recently abandoned troglodyte villages, to capture two prisoners, and to reach the summit of the northern side of the volcano, a point from which the whole of the Tibestian mountains can be seen.

The evenings, nights, and mornings were icy-cold, though the thermometer never fell below freezing-point. Our camels, taken aback by the novelty of the grass offered them, cropped it very sparsely; our provisions were giving out, and the rebels had fled before our arrival into exceptionally difficult mountainous tracts, where we could not dream of following them. In a word, in spite of the geographical interest there would have been in prolonging our stay on the summit of Emi Koussi, when the fourth day came we had to think about getting back to Yono.



STEEP SLOPES ON THE FLANK OF EMI KOUSSE,  
TIBESTI



NATURAL CISTERN OF DERSO AT THE FOOT  
OF EMI KOUSSI



THE GREAT CLIFF, TIBESTI



THE CRATER OF EMI KOUSSI (3400 m.), TIBESTI

From this excursion on the highest peak of the highest mountain in the Sahara I brought away an abiding impression of wild magnificence, and most of all when one's thoughts go back to the panorama of the Tibestian mountains. There may, I fear, be something of presumption in attempting even a short description; still, I will ask your permission to make a short extract from my diary on the day in question:

“ . . . Continuing our march northwards, we soon reach the foot of the cliffs of the northern wall, where, by a natural staircase, nearly 600 feet in height, one can reach the Tiribon pass, through which run the difficult paths that lead to Miski, Tozeur, and Goumeur.

“In front of us the volcano slopes steeply downwards, leaving open to view the Tibestian massif with the endless succession of points of its serrated ridges outlined against the sky and stretching away out of sight. On our left the crater-wall loses itself in a confused mass of rocks, while on the right rise a number of sharp peaks, one of which seems to be the culminating point of this part of the ring of heights that shut in the volcano.

“A last effort got us to the top of this lofty summit, 10,000 feet above the sea, where we found a narrow platform strewn with boulders, with big

clusters of red and lilac tinted flowers growing in the gaps between the stones. Toilsomely enough, I managed to scramble on to the highest rock, and as I stood on it, there lay before my eyes, for the first time, the mysterious Tibestian chains that no explorer had ever gazed on yet in their majestic entirety. The grandeur and beauty of the sight so far outdid all I had anticipated that I could not turn my eyes from watching the harmonious hues thrown over the landscape by the rays of the declining sun. The intense clearness of the air made it easy to see distinctly the remotest peaks; all around lay long ridges, their successive summits rising and falling in regular points like lace; scattered rocks, deep gorges, dizzy precipices, jagged peaks. Each mountain range, though all were turned by the sun to the purest rose colour, had its distinct shade, brightest in the foreground, softening into mauve as distance melted into distance away to the far horizon.

“Eastwards, the Tibestian massifs fell by giant steps whose sharp-angled lines, blurred by the first shadows of the waning day, ran into one another in inextricable tangles; while to the west the mountains bordered an endless plain, a forbidding waste of stones, over which brooded and deepened a gloom that threw into beautiful contrast the rosy-mantled chains whose lofty summits soared into a sky of calm and exquisite blue.”

Tearing myself away, not without reluctance, from the dreamy fancies called up by all these glories, I made haste to take a few observations with compass and thermometer and make a few notes. The Tibestian reliefs appeared to me to be included in a right angle, the apex of which is marked by the volcano, and the two sides by the directions W.N.W. and N.N.E.; such being the case, the appearance of Tibesti was totally different from what I had till then supposed it to be, on the strength of the statements put forward by the explorer Nachtigal. The rest of my journey was to afford me the opportunity of unravelling the skeins of the succession of ranges, whose apparent position and extent I could now approximately fix.

On September 18, towards noon, we struck camp, to go down again into the plain by the route we had followed on our upward march. While the camels, weary and emaciated, were painfully climbing the slopes of the pass leading out of the volcano, I took a last all-embracing look at this huge

crater, 10,000 feet above the sea; few others in the world are so immense, for it is 5 miles wide and 8 miles long, and looks like a gigantic funnel, almost elliptical in outline, 25 miles round and 800 yards deep; on all sides it is shut in by a rampart of unbroken wall, rising sheer almost everywhere for 500 or 600 feet, and which can be got over only at two points, by openings that are very hard to reach.

Behind this tremendous natural bulwark, 200 or 300 Koussadas live miserably, after the manner of cave-dwellers, divided into two clans, and possessing only a few camels, asses, and goats, and a small number of date palms in the neighbourhood of a few barely accessible springs dispersed here and there about the outer slopes of the volcano. Their staple food is a wild herb, the "Mouni," that grows among the rocks, and yields a coarse flour that looks like coal-dust; and in the plains at the foot of Emi Koussi they collect the seeds of a sort of bitter gourd, the "hamdal," which become eatable after undergoing a long preparation intended to take away their extremely bitter taste. At times they procure meat by hunting the "Meschi," a kind of wild sheep which is only to be met with in the high mountains, and of which throughout my journey I did not see a single specimen. They are supplied with stuffs, arms, and ammunition by the Senoussists of Koufra, to whom, profiting by the cool season, they bring goats in exchange; but the greater part of their scanty resources comes from the brigandage they practised until quite recently, with more or less success, on the routes that lead from Kanem to Borkou and Bilma. Untiring on the look-out, though not particularly brave fighters, they succeeded in keeping up an unrelenting watch on our movements during our exploration, and in this way they were able to get possession of one of our camels, too tired to keep up with us when we came down again towards the pasture-land of Yono.

We got back to our bivouac on September 20, and I had to stay there nearly a week to let the camels recuperate and to give them time to get better of the wounds to their feet caused by the sharp edges of the boulders they had had to walk on during that expedition.

I spent the week's rest in making calculations drawn from my different observations, and in exploring the hot springs of Yi-Erta, highly esteemed

in the whole region for their medicinal virtues. Their temperature is  $100.5^{\circ}$  Fahr. ( $38.1^{\circ}$  Cent.), and their flow of water by no means abundant. They can only be approached on foot and by a difficult path, in about an hour: their altitude is 3100 feet above the sea.

*Central Tibesti.*—When our camels had had a rest and feed in the pasture-lands of Yono, I decided to transfer my quarters to the great valley of Miski, 100 miles further north, skirting the western foot of Emi Koussi. This valley of Miski is one of the most important of the Tibestian massif, not in the matter of its alimentary products, which hardly exist, but from a military point of view, for the Tibestian rebels use it as a convenient meeting-place from which—with no great difficulty and without our knowledge—they can attack our southern and western lines of communication. In the course of our march (between 25 September and 1 October 1915) our patrols had a few small engagements with the rebels, and some prisoners were taken who supplied us with useful information: the Toubous, informed that our expedition was on the march, were gathering their crop of dates—though the dates were not fully ripe—and meant to seek refuge 100 miles further north-east, in the Tarso of Ouri.

The pasture-lands of Miski were already abandoned by the rebels, and so we were able to march without fighting through the two long passes that command the entrance to the valley. A number of reconnoitring patrols showed us the exactitude of the information mentioned above, except in respect of the palm plantation of Modra, where Lieut. Fouché's detachment, consisting of only fifteen men, had to put up a pretty hard fight in order to avoid being surrounded and cut to pieces.

The scarcity of food and the jaded condition of part of my camels forced me at this point to divide my forces and send part of them back to Borkou, after planning a new route. I remained alone with my secretary and thirty black soldiers to go on with my exploration of the heart of the unknown Tibesti. My aim was to effect a junction with the troops of Bardai in the valley of Yebbi, and to explore the gorges of Kozen and Goumeur in the east of the massif, where several rebellious tribes had taken refuge.

I left Miski on October 4, and on the 6th I reached the watershed between the basins of the Chad and the Mediterranean. At sunset I reached

the Mohi pass, 5000 feet high, but the gathering darkness prevented me making as good use (topographically speaking) of my presence at this spot as I should have been able to do if I had arrived there in full daylight. In that case, I might have climbed a commanding height of apparently easy ascent situated 2 or 3 miles east of the pass, from which position I should have been able to grasp the general character of this orographic centre. As it was, I had to cover the few miles that lay between us and the palm plantations of Yebbi in complete darkness, partly in the evening, and partly on the following morning. But through a mistake made by the guide it was only at half-past six that we saw the first palm tree, at the bottom of a dark valley shut in between almost vertical walls from 700 to 1500 feet high. The landscape on every side was inky black and beyond all expression desolate; the valley was covered with dark boulders, glistening in the sun; no trace of green could be seen, except two thin lines of palms bordering a stagnant watercourse hardly a dozen yards wide. High mountains were visible to the east, rising (so far as I could judge) to 6000 or 7000 feet.

To get down to the bottom of the valley there was only a narrow track littered with sharp blocks, on which our camels did not know where to set their feet. The vanguard that covered our toilsome descent was already exchanging shots with the Toubous, but was finally able to get possession of the palm grove; towards 9 o'clock we could pitch our tents, with no more fighting to do. A few goats and donkeys were our only booty. But soon there appeared three prisoners, almost naked, whose pitiable physical condition was strangely in keeping with the appalling wretchedness of a landscape that one might have taken for a vision of hell. They were miserable slaves, stolen by the Toubous during their forays against the inhabitants of Kanem and Wadai. Their state of mind was no better than that of their bodies, and there was little to be got out of them about the country and its inhabitants. At any rate, they enabled us to unearth a few hiding-places where we found some dates, a great boon to the members of the expedition, whose rations were growing daily shorter.

Towards 11 o'clock a Toubou envoy came, sent by the rebels to make terms for their submission; I offered very easy ones, and treated them with consideration. After half an hour's interview, I sent him back to the rebels on whose behalf he had come, but waited in vain for his return till evening.

Towards five in the afternoon I struck camp to seek a bivouac for the night, in a better position than the death-trap where we had spent the afternoon, and we halted, in complete darkness and without lighting fires, on a rocky platform that gave us 300 or 400 yards of open ground to fire over on all sides. Thanks to these measures, we were able to spend the rest of the night in peace.

Next day we went a little further down the valley in search of pasturage for our camels, worn out with hunger and fatigue; their condition left small hope of undertaking the excursion I had planned in the direction of Kozen and Goumeur, from which we were still separated by two or three ridges very difficult to cross, and where—so at least our prisoners said—neither pasture nor water could be found in readily accessible situations. When it is added that I had no news of the Bardai detachment which I had hoped to meet there, it will be understood that I thought best to advance in its direction two days' march further west, into the valley of Zoumri, where I was informed of the presence of friendly tribes who could probably supply me with some information about its movements.

These two marches were very hard on our animals. To cross from one valley to the other we had to make our way up a wearisome succession of ravines and steep slopes, one of which, on the sides of a spur of a precipitous cliff, cost the detachment a hard piece of work in making a flight of rough steps up which the camels, though completely unloaded, had the utmost difficulty in climbing. On the other hand, I had the good luck to see before me, on the east and north-east, a vast horizon of mountains which extended and confirmed the observations made on the summit of Emi Koussi, and made certain that the Tibestian massif, far from being limited to the simple mountain chain hitherto marked on the maps of Africa, stretched away for more than 100 miles into the interior of the Lybian desert. During the two hours required for the hard climb up this cliff I kept on taking observations of the numerous summits visible in the limpid distances of that ocean of rocks, summits that seemed to rise like a succession of landmarks along each of two or three long ridges in sharp and jagged peaks, equal in bulk and perhaps in height with those of the great western chain, of which a few outlines appeared in the gaps between the nearer ranges. But in face of this accumulation of lofty peaks I felt a bitter vexation, a sort of resentment

against my own littleness and powerlessness to set in order their apparent chaos. For it would have needed many a long excursion made with two or three fresh camel-trains, and a further provision of supplies, to enable me to straighten out the seeming tangle of these valleys and the confusing intersection of the hills.

Towards eight o'clock in the morning we resumed our westward march, skirting on the north an isolated mountain more than 8000 feet high, the Toh de Zoumri, which by its conical outline and the circular shape of its top looks like an old volcano, a supposition I had not time to verify. Our route crossed numerous tracks converging towards the mountains, which were used as a refuge by large numbers of Têda rebels, subjects of the former Dordëi of Bardai, whose revolt was aided by the encouragement and the supplies of arms and ammunition furnished by the Turco-Senoussists. Next day, October 11, we entered the valley of Zoumri by a pass 4800 feet high, and towards ten o'clock we bivouacked near the palm plantation of Yountiou, where I was hoping to meet with friendly Têdas who would put me in touch with the commander of the Bardai post. Unfortunately the village was deserted.

This fresh disappointment caused me little or no surprise; I expected my coming to Miski and thence to Yebbi to be known by all the hillmen, and that our skirmishes with the rebels would have been related with no small exaggeration as mighty combats; still, I felt that I was too near the goal to give up the attempt to reach it, so I sent out patrols to scour the neighbourhood and especially to capture a few Têdas who could guide me towards Bardai. Presently an old woman was brought to me, gaunt, stooping, and half crippled, but with intelligent eyes. After long reticence she confided to me that she was the mother of the chief of that village, and that her son had gone over to the French a few weeks earlier. Messengers had come during the two preceding days, announcing the coming of an expedition from Borkou, and when that morning the watchers saw our camels at the summit of the pass, all the Têdas—men, women, and children—fled panic-stricken into the neighbouring rocks; she alone had remained hidden in the palm plantation, because she said she was too feeble to follow them and too old to be afraid of death. I calmed her fears about my intentions as best I could, telling her that all the Têdas who submitted to

French authority could count on my good will, and urging her to bring me her son as soon as she could, promising her that she should be treated with friendship and consideration; but as I had to continue my journey to Bardai as soon as possible, she must understand that I should be obliged to procure guides by force if I could not get them otherwise. "You shall have a guide to take you to Bardai," she said, "and, if it please Allah, without needing to use your guns; I will go and tell my son." Soon after there came up a little man with the same intelligent eyes, young and timid looking. He handed me the certificate of submission given him only a few days before by the officer commanding the French forces in Tibesti. After a fairly long talk he declared himself ready to serve me, but begged me not to insist on trying to get any other men of his village, for they were grimly determined to stay in their hiding-places. I trusted him, and was rewarded for doing so, for he stayed at my disposition upwards of a week, and thanks to his knowledge of the country I was able to go on with my exploration as rapidly as possible, and to collect interesting geographical information about the regions that lay off the track of my journey. To go to Bardai we had only to follow the sandy bed of the dried-up river, along which from time to time we passed by palm plantations and villages, the headmen of which came to bid me welcome, pleading their poverty as an excuse for not offering me the customary presents. After twelve hours' march, when I had just passed through the village of Zoui, I met Lieut. Blaizot, commanding the troops of Tibesti, coming on foot to meet and welcome me and to express his regret that he had not been able, for want of camels, to come to Zoumri and Yebbi to help me against the rebels. To see him and to listen to his voice as he spoke were a great joy to me. In spite of all difficulties, I had just effected the junction so long desired between the troops of Borkou and those of Tibesti; in a few more minutes I was going at last to enter the palm plantation of Bardai that I had been dreaming of seeing for twenty years, ever since I had read in Nachtigal's impressive story of his travels about the difficulties he had to get over in order to enter it forty-six years before, and above all to get out of it alive. On the way I had been able to make a mass of observations, topographical, geodetic, and hypsometric, and to fix with a very satisfactory degree of precision the situation and height of the chief summits of the great western chain that Nachtigal had only been able to locate by guesswork, and often without having even seen them.

At Bardai, where I arrived on October 13 a little before noon, I stayed only twenty-four hours, for I was in a hurry to get back to Miski, where the little detachment left in charge of the broken-down camels and of my last reserves of food must have been in a situation of some insecurity since the 10th. During the afternoon of the 13th I was able to examine in detail with the commander of the garrison the various questions regarding the means of combining the efforts of the troops of Borkou and those of the Tibesti against the rebels. The night having been favourable to my astronomical observations and the morning to measurements of angles on the principal peaks visible from Bardai, I had been able in that short space of time to collect all the essential elements needed for fixing on the map with satisfactory exactitude the position of the most important points of Central Tibesti.

The geographical interest of my journey to Bardai did not consist solely in the discovery, to the east of the great chain traversed by Nachtigal, of mountains whose existence had not previously been suspected; it was greatly enhanced by the fact that my observations corrected serious errors of position and altitude committed by the famous German explorer on the itinerary he followed amid so many hardships. Thus, for example, in the site of Bardai there is an error of 50 miles in latitude and 30 in longitude; it is nearer 3000 than 2500 feet above sea-level; the height of the peaks of Toussidé and Timi is as much as 10,000 feet; the name of Tarso, which Nachtigal restricts to the massif he traversed, is a general term applied by the Tibestians to all mountainous regions consisting of high plateaux difficult of access, but on which the going is easy when once one has climbed to the top. Lastly, to the east of Bardai, instead of the great zone of plains shown on the maps there lies a succession of important massifs the culminating point of which rises as high as 8000 feet above the sea.

Refusing, albeit with extreme reluctance, to listen to the urgent insistence of my amiable host Lieut. Blaizot, I left the post of Bardai on the evening of October 14, and by a moonlight march lasting almost all night I was able to get back on the 15th to my bivouac at Yountiou to make the observations, astronomical and other, requisite for checking those of the previous days; from that point I counted on returning to Miski, not by the already reconnoitred route passing through Yebbi, but by the Modra route

lying further west, which was to afford me the opportunity of reconnoitring another passage. But a piece of news had just come which very much upset my Têda guide Mohammed: there had been fighting in the Modra valley between the Borkou troops and the hillmen, and he had very little fancy for guiding me through that region, where my detachment would presumably have to fight its way by main force. For me, on the contrary, it was a further reason for insisting on going there with all speed, in order to afford my companions, if need was, the help of the thirty rifles of my detachment.

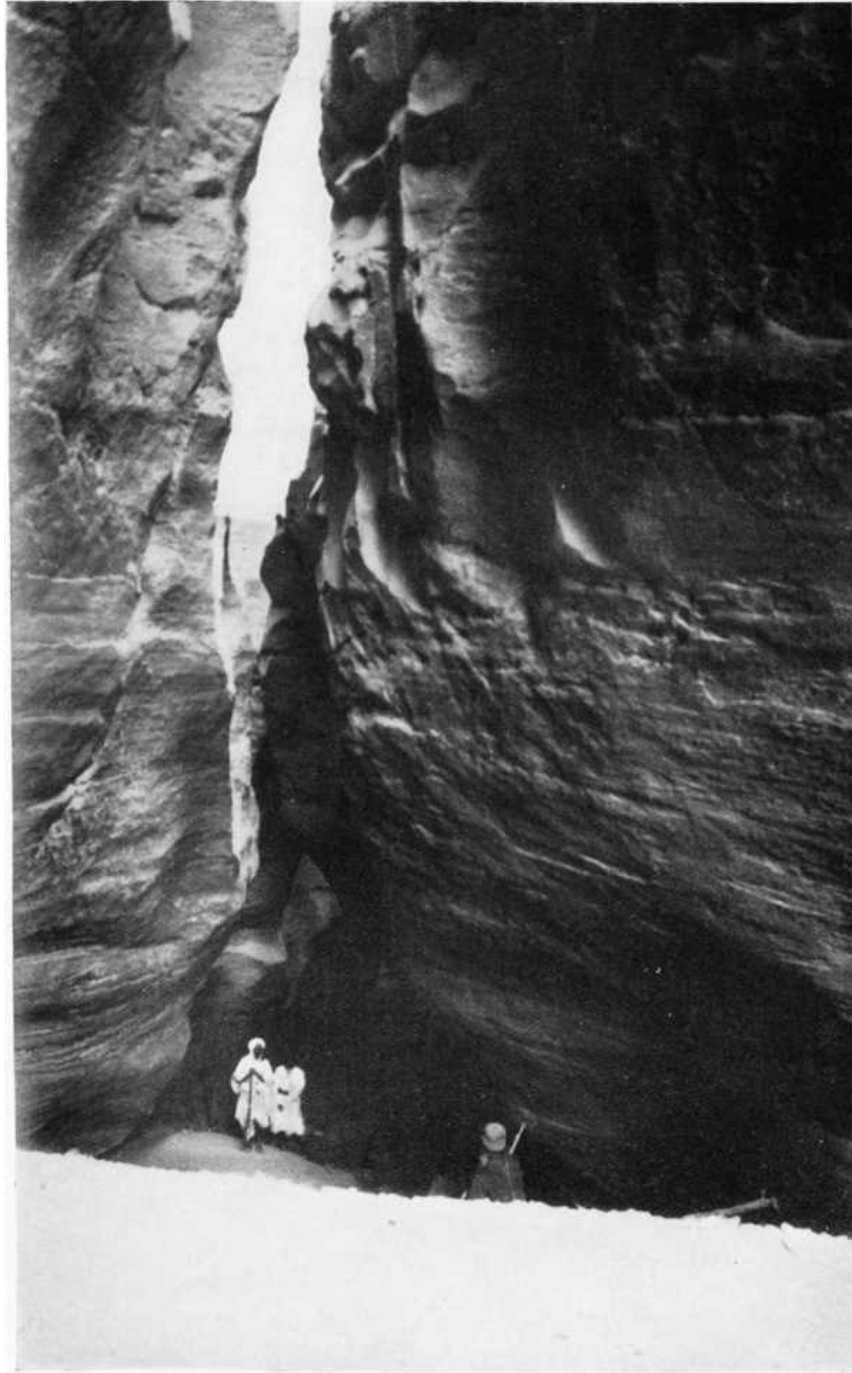
Mohammed allowed himself to be convinced by the promise of a suitable reward, and by the use of certain outer and visible signs indicating clearly that he did not guide me of his own free will: he adjusted a cord loosely round his neck, and one of my black soldiers seized hold of the other end. In the eyes of his own people his Têda honour was safe, and his responsibility for the consequences of the subsequent proceedings reduced to vanishing-point.

Mohammed guided us to perfection; the chain was crossed on the second day by the pass of Kidomma at an altitude of more than 6000 feet, and on the evening of the third day, after a very tiring march, we reached the point where the track leaves the plateau to go down into the bottom of the Modra valley. We got down a first drop of some 60 yards without very much trouble, in spite of the quarters of sharp-edged rock that rolled under the hesitating feet of our camels. Then, after perhaps a third of a mile of almost level going, I suddenly came in sight of the palm plantation of Modra lying at the bottom of a dark narrow gorge deep sunken between two almost vertical walls more than 1500 feet high.

I was not without uneasiness at this sight, and came within a very little of thinking that the worthy Mohammed had deliberately lured me into some trap when he had said to me: "The descent into the Modra valley is rather difficult, but good camels can get down." The descent into the valley of Yebbi, which I had found so arduous eleven days previously, seemed to me now quite a reasonable sort of descent compared with this one. Already the valley was echoing with the reports of rifles; here and there I saw Toubous climbing the cliff-sides like goats and stopping now and then to favour us

from afar with noisy but harmless shots, and vigorous volleys of bad language more harmless still.

There being no conceivable alternative to consider we had to go forward. Covered by an advanced guard that returned the Toubous' fire with a fusillade of doubtful efficacy, and by a rear-guard that watched the points from which the rebels could have rolled down tons of rock on our heads, we crawled downwards in a circumspect advance along a path that was no path—that clung to the face of a steep cliff, now plunging sharply downwards in short zigzags, now hanging, a narrow ledge, above the abyss towards which great stones dislodged by our camels rolled rumbling or leapt clattering down from tier to tier. The camels were frightened; they had to be led forward one by one, and could only be got round corners with many stripes and voluble cursing. A little group of men went ahead of them, thrusting aside the most awkward blocks, and, where the natural steps in the rock were too steep, laying flat stones at the foot so as to break them in two. The descent was so toilsome and so slow that at sunset we were only halfway down. I had to call a halt, profiting by a little rocky spur that afforded us a narrow rugged platform where we found just room enough to make our camels kneel and to install our bivouac. The firing had almost ceased: our advanced guard came in soon afterwards after forcing the rebels to abandon their villages, the conical roofs of which could be seen shining in the moonlight more than 400 feet below. Still further down, below the palms, ran an invisible stream, forming a monotonous waterfall that we heard murmur in the neighbouring rocks.



A WATER-HOLE IN TIBESTI



FIRST BUTTRESSES OF THE MASSIF OF TIBESTI

Above our heads little patrols, relieved from hour to hour, kept watch on the upper slopes from which the Toubous might have sent undesirable avalanches rolling into our camp. The narrow band of sky that we could see was filled with shining stars, by which I could make the observations needed for calculating the point where we had stopped. The night passed, calm and silent, and next morning, after an hour and a half of fresh efforts, we were able to take up our quarters quietly on the banks of the stream.

After which the excellent Mohammed, having received the promised reward, took leave of us to return to his palm grove at Yountiou. But his prudence led him to take quite another route, accessible only to men and goats. All the luggage he carried was a little skin bottle half full of water hanging from his right shoulder, together with a tiny bag containing a few handfuls of dates and about a pound of millet flour. On his left shoulder, swinging triumphantly from the two ends of his staff, were two fine large-sized biscuit tins that glittered in the sun and resounded like beaten gongs whenever they knocked against the corner of a rock.

Toubous in small numbers still showed themselves on the cliff-sides, but did not wait for the patrols I sent to parley with them. After a few hours

spent in watering the camels and in filling our barrels and skin bottles, we resumed our route towards Miski. The little river of Modra ran hardly more than a mile further down the valley, and the dry bed of the torrent, at first littered with boulders, soon turned into a fine winding road of sand from 200 to 300 yards wide. Twenty miles further on we had to leave the river-bed and plunge into a chaos of little ridges of schist, intersected by narrow valley-ways leading into valleys that came down from neighbouring high mountains of an altitude exceeding 9000 feet: our camels had much trouble in making headway among sharp edges of slaty rock upturned almost vertically. They zigzagged from pass to pass, climbing steep slopes, dropping into rocky ravines, beyond which fresh ridges separated by fresh ravines rose in endless succession. At last on the 21st, very early in the morning, we came out into the wide flat valley of Miski, where we made a brief halt to allow the stragglers to come in. All our camels were there except one, and I may say that I felt much satisfaction at having succeeded in bringing them back to the starting-point after this toilsome flying expedition of more than 300 miles, carried out in seventeen days in the unknown and exceptionally difficult mountain region of which I have tried to give you as closely exact a description as I can.

For another 15 miles we pursued our way in the great valley of Miski, of an average width of 4 to 5 miles, finding it pleasant to look once more on the well-known landscape of peaks, domes, and cliffs of the Tarso Koussi. The clearness of the air was such that all these mountains seemed to be within walking distance, and that in this vast bare basin where not a breath of air stirred and where the sun blazed his hottest, we had the impression of marching without making any progress, so unchanging did the perspective remain.

Towards 10 o'clock we found the first siwak bushes with their characteristic peppery smell, and clumps of hamal, or bitter melon, with their dried-up fruits; then, a little further on, a few stunted and scattered talhas, a sort of acacia. At noon I got back at last to the bivouac where my secretary was waiting for me. For five days, since the departure for Borkou of Lieut. Fouché's detachment, he had been left alone with seven soldiers and seven camel-drivers to guard the supplies and the reserve camels. And when I asked him whether the Toubous had not worried him during that

spell of isolation, he showed me his zeriba, well organized for defence, with cartridge-boxes ready opened, and replied sadly, "No such luck."

To console him for his long inactivity I put him in charge of a patrol sent against Youdou, a palm plantation still held by rebels, and of which the site was not known; but he had not the good fortune of coming to grips with them, for the alarm was given by their sentries, and they drew off northwards into a rocky country where we should have had much difficulty and lost a great deal of time in pursuing them. None the less, this rush of 80 miles in less than forty hours across the awkward country of the Tarso Koussi foothills achieved its purpose of forcing the rebels to withdraw and fixing the site of Youdou with the desired precision.

*Western Tibesti.*—Thus the most important part of my geographical and military programme in the Tibesti was carried to an end; at no point had the Toubous offered a serious resistance to our march, in spite of the magnificent defensive positions their country afforded them. The most unruly among them had fled away to the north-east, more anxious to get to a safe distance than to carry out their aggressive schemes against our convoys of supplies; the rest, beaten off at every encounter, had let us explore their wild valleys without subjecting us to any surprises, whether in the shape of ambushes or of the capture of camels in grazing-time. Lastly, the general physiognomy of the Tibestian massif was revealed with sufficient clearness by my various observations, and its real position determined with all desirable precision. It only remained, before returning to Borkou, to explore the valleys of the western slope, and try to form a junction with the camel corps of Zouar.

I accordingly set out for Tottous, an important water point 70 miles further west, in the Wadi Domar where it comes out of the last foothills of the Tibesti. The distance was covered in four days with little trouble by following the lower valley of the Wad Miski, of which I was thus enabled to cross in succession all the tributaries on the right bank, till then unknown. The officer in command of the Zouar camel corps, having been informed after my visit to Bardai that I was desirous of seeing him, came to meet me, and we reached Tottous on the same day. He was accompanied by the chief of the Tomagras, the noblest tribe among the Têda-tous, the aged Guetty,

who had made his submission to the French authorities a few months earlier. Guetty was a handsome old man with a white beard and a skin less dark than usual. He was tall and regular featured, but his keen sly face inspired me with no great confidence; he was suspected of double-dealing, and of supplying the rebels with fuller information about our movements than us about theirs. During two days we had long conversations about the restitution to their families of the women and children that his fellow-tribesmen had carried off in 1913 in the course of a razzia on an Arab tribe of Kanem; but the old rascal either could not or would not fall in with my wishes, declaring truly or falsely that the luckless captives had been sold as slaves and sent away for the most part to the Senoussists of Cyrenaica.

*The Return Journey to Borkou.*—The exhaustion of my camels had reached such a point that I had to stay five days in the grazing-grounds of Tottous. I profited by the delay to explore the course of the Wadi Domar for about a score of miles in company of the Zouar camel corps, who were going back to their station. My food supplies, which had not been renewed for two months, were coming to an end, and I could not further prolong my excursions in the valleys of Tibesti. Besides, the greater part of the rebels had concentrated in the region of Abo, at the north-western end of the massif, twelve whole days' march away from Tottous.

Starting on November 4 for Faya, by a route hitherto unreconnoitred, we covered 120 miles of desert in six days before reaching the oasis of Kirdimi, near Ain Galakka, by the last and utmost effort our camels were capable of. On November 12 at nightfall I found myself back in my post of Faya, whose stout clay huts seemed to me for a whole week afterwards, if not absolutely the last word, at least the last word but one of comfort and civilization in the heart of the Sahara.

## **8. Military Operations in 1916-1917.**

This exploration of Tibesti marked the end of the long journeys that had been indispensable to the acquisition of a general knowledge of the vast desert regions placed under my authority. The calculation of my numerous observations, the making of general maps, the setting in order of my notes of travel, and the writing of reports to be sent to the Government occupied all my leisure in 1916. There was not much of it, by the way, for distant effects of the world-war were already beginning to be felt in Africa. The Grand Senoussi, Ahmed Sherif, was lending a more and more willing ear to the suggestions of Nouri Bey's Turco-German mission, and sending one emissary after another to preach revolt to the different sultans responsible to the French and British authorities; his exhortations were particularly well received in Dar Four and in the south of Wadai, where the English Colonel Kelly and the French Colonel Hilaire had to do some serious fighting before they could restore order.

In the desert country I had charge of, the unrest had become almost general among the nomads, and my camel-corp patrols had hard work to maintain the regularity of our communications: there were rumours of a great expedition of Germans, Turks, and Senoussists, with cannon, machine-guns, and five thousand fighting troops, which was said to be forming at Koufra to cross the Libyan desert and drive the French from Borkou, Tibesti, and Ennedi. We made superb defensive preparations, but no expeditionary force from Koufra ever came; what did come to reinforce the rebels were brigands and highway robbers who made the roads unsafe, and whom we had to pursue in all directions more or less. Among the most remarkable of the expeditions of this period two deserve special mention: they were led by Adjutant Amboroko, an old black non-commissioned officer whose energy, courage, and high spirit won universal admiration.

Having received orders to go in pursuit of a strong party of Toubous commanded by Mohammed Erbeimi, a particularly dangerous leader of raiders who had just made a successful foray in British territory, he began

by covering 130 miles in three days. Then for four days he patrolled the neighbourhood of Tekro without being able to find any trace of his enemy. He learnt, however, that Mohammed Erbeimi was encamped 130 miles further east, and again covering that distance in three days, he reached the well of Bini Erdi only to find that the band had decamped two days earlier, following in the opposite direction a route nearly parallel to that by which he had come. Allowing his detachment just time enough to water their camels and fill their skin-bottles, he set out again at once, following the tracks of the raiders and forcing the pace! The pursuit, hotter and hotter as the trail of the rebels grew fresher, lasted fifty-one hours, two of which only were allowed for rest, and he came into contact with the rebels at dead of night. Unluckily, the barking of their dogs gave the alarm to the enemy at the last moment. Our men leapt down from their camels and made a sharp and sudden attack on the Toubous, who had not time to organize their defence and fled headlong into the neighbouring rocks, leaving on the ground four killed, all their camels, and the prisoners they had taken in Dar Four.

Some time afterwards Mohammed Erbeimi made an attempt to get his revenge. Reinforced by a contingent of Senoussists from Koufra, he organized a flying column a hundred rifles strong and flung it by a rapid march on our lines of communication between Borkou and Wadai, where our last supplies of the year were on their way. Thanks to the treachery of a Nakazza chief, he was able at daybreak to surprise one of our convoys on the march. Though the escort counted only fifteen rifles under a black sergeant, our black troops offered a bold front; but, overpowered by numbers and deserted by the camel-drivers, all they could do was to save their honour and fall in their tracks. That took place 150 miles south of Faya, in the desert of Mortcha. Now, it so happened that Adjutant Amboroko, with a force of seventy-five rifles, had been patrolling for two days in that same desert, on the look-out for Mohammed Erbeimi's raiding party, my spies having notified me, albeit rather late, of its appearance on the scene. He was not able to get on its tracks till sixteen hours after the wiping-out of the convoy escort, when he set off at once in pursuit. Two hours later he came upon it by surprise and routed it in a few minutes by a vigorous bayonet-charge; the enemy, taken completely off his guard,

abandoned his booty and a certain number of dead, and made off hastily eastwards. Amboroko, an old hand at desert fighting, thereupon judged it expedient to let the Toubous get a few miles' start, and so lead them to think that he held himself satisfied by the recapture of our supplies of cereals and of our camels, and was going to take back the camels at once to Faya. He calculated that as soon as the first spell of panic was over the rebels would get together to discuss the advisability of a counter-attack. His forecast turned out correct. Resuming the pursuit under cover of night, he again came in sight of the raiding-party towards three in the morning, in regular order once more, and holding a palaver round the bivouac fires. Closing in to short range he poured in a rapid fire, immediately followed by a bayonet-charge that laid out a dozen Toubous, while the rest in utter panic fled at top speed in all directions, some on foot, others hanging on to the tails of their camels that made off at full gallop without leaving time for their riders to get astride. The hunt went on till noon, and supplied us with a few prisoners who gave the most precise details of the treachery of the Nakazza chief; after which Amboroko retraced his steps to take in charge the convoy of supplies and bring it into Faya. But he was of opinion that our brave soldiers fallen the day before were not sufficiently avenged, and providing himself with fresh camels he set out at once in pursuit, seeking all across the desert the tracks of those who had escaped his two counter-attacks. Going further and further afield, he found himself finally 300 miles to the eastward among the rocks of Erdi, where the families of Mohammed Erbeimi's Toubous were in hiding, and engaged in two fights with them which cost the rebels some thirty killed; but the old chief unluckily succeeded once more in bringing his head safely out of the business.

Early in 1917 the revolt might be considered as crushed. The tribes had begun to discuss terms of submission, all except Mohammed Erbeimi's tribe, the remnant of which had taken refuge in the massif of Ouri 300 miles north-east of Faya, and was not in a condition to do any harm for a certain time.

## 9. Homeward Journey.

Then I saw my interminable sojourn in the desert brought to an end by the person of Captain Gauckler, an experienced commander of camel-corps, who had seen most of his service in the African colonies, and was come from the French front to replace me in Borkou. Thus my turn on the Western Front was to come early enough to enable me to share in the gigantic battle that could be foreseen, from the hour when Russia fell out of the fight, as imminent and decisive. The French Government having replied favourably to my request for permission to return to France by way of Egypt, this return journey would allow me to effect the geodetic and topographical liaison between Borkou and Dar Four—in other words, to accomplish the last part of the geographical programme that toward the end of the last century I had set myself to carry out.

*From Borkou to Wadai.*—I left the oasis of Faya on 25 April 1917 in an east-south-easterly direction, skirting the foot of the western spurs of the high tablelands of Ennedi. In ten days I reached the post of Fada, where Captain Châteauneuf presented to me the chiefs Gaëdas and Mourdias, whom two long years of incessant struggles had constrained to submit; we discussed and settled in concert the conditions on which the “aman” should be granted them. After which, turning my back on the picturesque rocks of Ennedi, I went on my way towards the south-west, across the desert of Mortcha, to reach the wells of Oum Chalouba. These wells, situated in the Wadi Hachim, belong to the Nakazzas, one of the principal Toubou tribes of Borkou, who are masters, under our control, of the oasis surrounding the post of Fada, but whose submission to our authority did not prevent them from entertaining with our enemies relations as cordial as they were clandestine, that gave us endless trouble. The judgment-seat of the native court over which I presided was heaped high with complaints and claims for damages against their chiefs, Allatchi and Djimmi. Their low cunning and double-dealing exasperated me; but since my return to Europe it has become evident to me that, like many other reputable persons, they were simply engaged in politics.



but they produce extensive and abundant pasturage, where Mahamid tribes graze fine herds of oxen and flocks of sheep and goats.

Two military posts ensure the policing and administration of the country: Arada, the commissariat centre of a camel-corps section, and Biltine, where a company of black troops is garrisoned. It is in the neighbourhood of Biltine that the first villages of the sedentary tribes are seen, the Mimis, then the Kodois. The millet fields, small at first and far apart, increase in size and frequency as one gets further south; but the harvests are still uncertain, for spells of drought are by no means rare. The year 1913 was especially fatal; the grain dried up on the stalk, and there was such a shortage when the crops were got in that a terrible famine spread over the whole country during the first eight months of 1914. Many inhabitants had to emigrate southwards, and those who had not foresight enough to flee in time, chiefly old men and children, died of hunger in the villages they had not been willing to leave. The number of the inhabitants of Wadai who perished thus is estimated at more than half, some say even at more than three-quarters. The population of Wadai, put by Nachtigal at more than two millions in 1872, had fallen to 300,000 when I went that way.

*Abéché.*—At sunrise on 31 May 1917 I came in sight of Abéché, the famous capital of the sultans who had made of Wadai one of the most powerful Soudanese kingdoms of the nineteenth century. Seen from a distance, it looks like a little cluster, grey and huddled, of low houses, overtopped by a few towers with pointed roofs, and had nothing of the handsome appearance that had impressed Nachtigal nearly fifty years before. It was now no more than a small town of three or four thousand people, and more than half ruined. It is true that ruins are accumulated with extreme rapidity in Central Africa, where the finest houses are only ill-built huts of clay kneaded and baked in the sun, and quickly falling into dilapidation every rainy season. The plain surrounding the town looks no better, being scantily covered with dry grasses and little green clumps of “m’keit” which our camels browsed on with lively satisfaction. The shrub-tribe was almost exclusively represented by little “oshar,” whose puffy-looking fruits enclose a silky down like “kapok”; as for the mimosa family, so abundant in the neighbouring bush, it had well-nigh disappeared, as often

happens near the negro habitations through the wasteful use made of it as firewood.

Abéché has retained few traces of its ancient splendour. The former palace of the sultans, kept till that time as a specimen of the architecture of Wadai, had just been pulled down by order of the new governor of the province. Round about it was strewn a mass of *débris*, on which were slowly rising new buildings of a highly military style. Only the business quarter of Am Sogou and the market-place had kept a busy and animated aspect. Men, women, and merry black small-fry bustled noisily to and fro, inextricably mixed up with asses, camels, dogs, and horses. Numerous Tripolitan merchants, white-faced, wearing red fezzes and long flowing embroidered robes, stalked gravely back and forth, making it evident by their decorous elegance and the satisfaction visible on their faces that, in spite of the suppression of the slave-traffic, business remained active and prosperous.

*From Wadai to Dar Four.*—I was forced, much against my will, to stay ten long days at Abéché before continuing my journey. The road usually followed from Abéché to El Fasher passes through Dar Massalit to Kebkebia, along the valleys of Wadi Kadja and Wadi Barré; it is about 220 miles long and very easy, except from August to October or November, when the summer rains fill the rivers and temporary marshes, very numerous in this region. But since that route had been reconnoitred formerly by Nachtigal, and very recently by Colonel Hilaire, the idea had occurred to me of studying a more northerly route unknown throughout two-thirds of its length, and passing through Dar Tama, Dar Guimer, and northern Dar Four.

*Dar Tama.*—This project having obtained the approbation of the Government, I was able to leave Abéché on June 9, and plunged into a very broken granitic region, where the rise and fall was inconsiderable, but which was intersected by numerous wooded valleys where marching was no very easy matter, especially at night. But I had the advantage of passing through an inhabited tract where water was frequently to be found, a consideration of importance for the feeding of a little group of Zagawa women and children whom I was taking back to Dar Four after a long and

eventful sojourn in the wilderness. Captured the year before by the same Toubou raiders whom we had to go in pursuit of, they had been delivered by our camel-corps, and were going back to their families under the protection of my escort. We went from village to village, forced to change guides at every halt, and to stay long enough to listen to the compliments with which the notabilities bade us welcome. In addition to the compliments, they brought us water, millet, eggs, a little milk, and sometimes a sheep or a goat. Around the villages there were many fields of millet and sorgho, and it was not unusual to meet with gardens, in which cotton, tobacco, and spices were the most frequent products.

In this way we reached the plateaux of Dar Tama, averaging from 2500 to 3000 feet in altitude, where on the gently undulating surface the going was pleasanter than on the rough slopes of the foothills leading up to the tableland. A few lonely eminences rose here and there, the loftiest of which, the peak of Niéré, visible for 30 miles around, reaches a height of 4500 feet. For the first time in more than four years I saw once again the thick-leaved tamarind trees, whose beautiful green is a rest to the eyes, and in whose shade the traveller is glad to halt during the hottest hours.

On June 13, after a long stage during which our successive guides had led us in needless zigzags, we arrived at the foot of Mount Niéré, where there is a village called Nannaoua. Here we camped in the deep shade of two or three white acacias, less than 500 yards from the spot where in 1909 one of the brilliant contemporary explorers of Central Africa, the regretted English Lieutenant Boyd Alexander, was assassinated. My tent had hardly been pitched an hour when a messenger came to announce the visit of the Sultan of Tama, who desired to present his compliments and bid me welcome. This mark of courteous deference was all the pleasanter from the fact that on leaving Abéché I had been put on my guard against a possible want of cordiality during my passage through Tama. I immediately had a mat of palm-fibre, in default of carpets, laid down at the entrance to my tent, and advanced to meet the sultan, a handsome, white-bearded old man with a black skin and kindly intelligent eyes; he was dressed in the flowing robe in use throughout Central Africa, but made of fine linen richly embroidered. He wore brown boots made in Europe, and his careful attention to his personal appearance went the length of socks. On his head

was a red fez, round which ran a narrow twist of white muslin, and he walked with slow and stately steps, his left hand resting on the shoulder of one of his servants.

Our interview lasted upwards of half an hour, and was extremely cordial; the sultan urged me to break up my camp the same afternoon in order to go and sleep in his capital of Niéré, where he had had huts made ready for us; but in reply I alleged the exhaustion of our camels, which were in urgent need of grazing till evening. Besides, I had to make a stellar observation at that particular spot in order to calculate exactly the position and altitude of the mountain of Niéré, the most remarkable point, geographically speaking, of the whole region. Soon afterwards I saw the sultan was waiting for me to rise and take leave; I helped him up and accompanied him a few steps from my tent. His servants and dependents were waiting outside for him in the ritual attitude of the courtiers of the ancient sultans of Central Africa, that is to say, prostrated to the ground, their knees and elbows resting on the earth, and their hind-quarters level with their head.

He called the chief of the village of Nannaoua to give him instructions with a view to our comfort. The latter got up and came to listen to his suzerain's commands, kneeling before him with clasped hands, downcast eyes, and devoutly attentive face. When the sultan ceased speaking, the village chief clapped his hands several times and got up to go at once and transmit to his subjects the orders he had just received.

Early next morning I reached the camp that had been prepared for me in the shade of some "kournas" near the well, but the huts were so low roofed and uncomfortable that I preferred to pitch my tent, severely damaged as it was by four years' wear and tear. I had to stay two days at Niéré to wait for the arrival of four camels intended to replace the pack-carrying oxen I had to send back to Abéché.

The capital of Tama is only a small village covering about 35 acres, where the straw huts are set rather far apart; the inhabitants, by no means numerous, consist almost exclusively of the families and servants of the dignitaries immediately surrounding the sultan. Other villages are scattered about the neighbourhood, usually lying at the foot of isolated rocks of no

great height, but of very characteristic geometrical shapes, rising out of the uniform tableland like natural landmarks destined to rejoice the hearts of a triangulation brigade.

In our camp an unpleasant surprise awaited us: hardly had we settled down when we saw coming down from the kournas whole battalions of caterpillars that made straight towards us and obstinately set about climbing all over our packing-cases, chairs, clothes, and persons in quest of a quiet and shady corner where they could comfortably instal their cocoons and go to sleep in the hope of a happy metamorphosis. We hunted them, killed them, but to no purpose, for still they came. And these caterpillars, sociable to a fault, are tormentors of the worst type: wherever they go they leave behind them invisible hairs that burn like nettles. Next morning we were all scratching furiously, unable to find even momentary relief except in applications of very hot water. My trunk of books was infested, and, above all, that which contained my linen; so also were my bedclothes. All the washing, swilling, and beating I could do failed to rid my clothes entirely of this pest, and I had to endure its tortures for long as best I might. It was only when I got to Khartoum and could get fresh clothes and throw away my up-country garments, if such they could be called, that I really found a little peace. In the evening a thick cloud of locusts came and settled on the region; in a few minutes the trees were covered with them, and their green changed to the pink hue of these voracious insects' bodies.

The sultan came repeatedly to see me. He was fond of talking and telling me his history and that of Tama during the preceding decade; he also told me the story of the murder of Boyd Alexander as it was related to him not many days after the tragic event by his predecessor the Sultan Othman and the chief Adem Rouyal, commander of the Forian force sent from Dar Four by the Sultan Ali Dinar to drive the French out of Wadai.<sup>[2]</sup> The sultan was above all interested in the Franco-Anglo-German war; he asked question after question, and I had a great deal of trouble in giving him a hazy idea of the formidable masses of war material, supplies, cannon, rifles, and the unheard-of numbers of men brought into action on both sides.

Thanks to his good offices, I was able to get the supplies I was in daily need of for my detachment; and in these days of excessively dear living it

will not perhaps be without interest to give a summary list, at this point, of the prices that were asked me:

|                          | s. | d. |
|--------------------------|----|----|
| A small yearling ox      | 12 | 0  |
| 200 lbs. of millet flour | 4  | 0  |
| An average-sized sheep   | 2  | 6  |
| Chickens                 | 0  | 6½ |
| One pound of butter      | 0  | 3  |
| „ „ onions               | 0  | 3  |
| A quart of milk          | 0  | 1  |

Had we been wise enough to have rational ideas about railways in Africa, and to have them in time, what a help the Black Continent would be to us now! I trust the ordeal we are going through to-day may induce France and Great Britain, the two great guardians of the Black population, to join in intimate union in order to labour together at the great work of opening up Africa and turning its resources to account—a work that must be undertaken at once! But this is a vast question, and one that must be treated separately; so I beg to be excused for this digression.

In the afternoon of the 10th, having succeeded in hiring the necessary five camels, two of them enormous, and the other three of the tiniest, I took leave of Sultan Hassan to go on with my journey towards Guimer. Four days later I arrived at Koulbouss, the temporary residence of the Sultan of Guimer.

*Dar Guimer.*—The welcome I received was of the chilliest. Two hundred yards from the village a son of the Sultan Idriss came all alone to meet me, and announced that his father had started a few days earlier for El Fasher; and then, skirting the village, he led me down the valley to a spot where a dilapidated hut, not far from a well and at the entrance of what had once been a piece of enclosed land, was offered me in which to take up my quarters. I had great difficulty in obtaining a few provisions, and two days were spent in animated discussions before I could get a guide and four hired camels to replace those lent me in Tama. Even so I only got them thanks to the good offices of a Zagawa chief who had come to greet me on my

passage because he had on a former occasion found his relations with the French authorities of Wadai turn out greatly to his advantage. But I could not get the sort of current information about the country and its inhabitants usually given to travellers by the natives. However, when I showed my surprise at the residence of the Sultan of Guimer at Koulbouss, which is in Tama territory, the son of Sultan Idriss condescended to explain that that installation was only temporary, having been authorized towards 1910 by Sultan Hassan of Tama by reason of the raids the Sultan of Guimer had had to undergo at the hands of the Forian bands of Ali Dinar. His return to his own capital was to take place shortly, the occupation of El Fasher by the Anglo-Egyptian troops having put an end to these incursions.

I left Koulbouss on 22 June early in the morning, with no great confidence in the success of my enterprise, for the guide assigned to me did not seem any too satisfied at the idea of taking me to Kebkebia, from which we were separated by a stretch of almost completely uninhabited country nearly 120 miles across, and in which the water-points were few and quite possibly dried up. Very luckily, everything went as well as could be imagined; I saw no trace of the Senoussist raid, so called, which local rumour credited for some time with having caught me by surprise, taken me prisoner, and carried me off as a hostage to Koufra. A few wells were found, very nearly dry, but we were careful in husbanding our supply of water. We saw very few inhabitants and met no caravan. What worried me most, and most unexpectedly, was the grazing question, for the country, though covered with scrub, was so dried up that our camels hardly ever got a satisfying feed and grew most disquietingly thin.

Dar Guimer is hardly more than a gently undulating plain of somewhat uniform appearance, 100 miles across from east to west, and 20 from north to south. The inhabitants, few in number, if I may accept the accounts given me, seem less inclined to tillage than to cattle-raising. The soil is usually clayey, very marshy from the end of July to December, but almost completely waterless from April to July. The valleys come down fanwise from the tablelands of Tama on the west, of Zagawa on the north, and northern Dar Four on the east. They meet on a level with the Djebel Kichkich (Hadjer Moull) to form the Wadi Kadja, one of the parent

branches of the Bahr-Salamat, which is one of the most important valleys on the right bank of the Shari, the main affluent of the Chad.

During the morning of June 25 we reached the southern limit of Dar Guimer at the wells of Taziriba; only 3 yards deep and flowing abundantly at all seasons, they were situated in a valley where there are no trees of any size, but an abundant growth of scrub. The wells, usually silted up, had been dug out afresh a few days previously, on the occasion of the Sultan Idriss' visit to Dar Four. Having thus been able to water our camels and renew our own supply, we left the territory of Guimer the same evening, to go and sleep half a score of miles further on.

*Between Guimer and Dar Four.*—It is interesting to notice that the tribes whose territories separate Wadai from Dar Four (Massalit, Tama, and Guimer) have always left a wide belt of uninhabited country between themselves and Dar Four. At some points its width exceeds 100 miles, while no similar solution of continuity exists between them and Wadai. It should not be concluded, as is sometimes done, that these territories are desert-like in character, for they are watered every year by the summer rains and covered with an abundant vegetation, for the most part thorny and stunted, it is true. These lands are not incapable even of settled habitation, for it would suffice to bore a few wells, around each of which men could take up their quarters in permanence, with fields of grain and cotton and pasturage for cattle. Such unpeopled regions are common in Central Africa, and each of them constitutes a neutral zone, a sort of “no man's land” that separates the territories of two hostile tribes.

It was across a belt of this kind that our route now lay, a belt about 70 miles wide between Safé, the last village of Guimer, and Rémélé, the first of Dar Four. On June 26 a long morning march brought us to the wells of Délébé, situated at the crossing of an important route chiefly used by native traffickers on their way to barter the grain of Massalit for the salt of Dar Four at the market of Diellé, some 20 miles north of Kebkebia. The site was pleasant and covered for a space of several miles in length and 200 or 300 yards in breadth with fine harazes and kournas, which gave us the illusion of a great shady park at home; but the lack of water in the well and the way

our store of eatables was running short did not allow us to yield to the temptation of resting there a day.

We had to start again in the afternoon and march till dark in order to reach, early next morning, the wells of Chibéké, whose immediate neighbourhood, so our guide told us, was infested by lions; but we had not the pleasure of seeing any. A further stage of a score of miles at last permitted us to get out of the uninhabited region and reach the Wadi Gueddara, at the point where it comes out of the mountains that mark the watershed between the basins of the Chad and the Nile.

*Western Dar Four.*—These mountains seemed to be much more important than the maps and descriptions of former travellers had led me to suppose. They formed a long and rather confused chain, running approximately from north to south; and their chief summit, mount Dourboulé, some 30 miles to the east, rose to more than 7000 feet above sea-level.

I spent June 28 at the village of Rémélé, where I received a very kind letter of welcome from Lieut.-Colonel Savile Pasha, governor of the province, who put at my disposal an escort of six soldiers of the native police. I wanted to ascertain the exact position of this village, but rain fell at intervals throughout the evening and night and prevented me from observing the indispensable stars. If I was vexed, the natives were delighted, for the damp soil would enable them to sow seed for the first time that year. Next day I had only a dozen miles to cover in order to arrive at the advanced post of Kebkebia, the furthest west of the military posts in Dar Four, and during that short march I enjoyed the happy and restful feeling of the sailor who, after a long voyage, sees shining on the horizon, across the calm of the spent waters, the cheerful harbour lights. We advanced along the western foot of the chain, gradually nearing it, and noticing that it seemed to connect with the massif of Djebel Marra, of which from time to time I could see for a moment the highest peak, more than 50 miles to the south-south-east. We went along through a smiling and prosperous-looking country, already covered with springing grass, dotted with green trees, and broken here and there by rocky heights that did not rise higher than 400 feet.

The natives, scattered about their fields, watched our caravan go by without unfriendliness or sign of misgiving, and then betook themselves again to their work with the serene dignity of men who till the soil. Both in the explicit picture it makes and in suggestion, their husbandry is very different from ours. The noble gesture familiar in our western fields, of the sower sowing his seed broadcast along the furrows, is lacking on African plains. The man I was watching walked straight on, holding in both hands a hoe bent into a right angle; at every second step, without stopping or even stooping, he made with it a tiny hole, hardly more than a scratch in the tawny sand. He was followed by a child, a boy clad in a simple sunbeam, carrying a calabash of millet, and parsimoniously letting fall into each hole a few grains that he summarily covered by turning a little earth over them with his bare toes. Happy lands, where man is satisfied with hard, coarse grain, and where the earth, in return for but small pains, breaks forth into abundant harvest. Which of us shall judge between them, and say whether it is better to be exacting in one's wants, and with great labour to attain to one's desire, or to be content with little and find that, with hardly an effort, that little may be had?

I was welcomed on my arrival at Kebkebia by the commander, a native officer of the 13th Sudanese Battalion, Sub-Lieut. Saïd Effendi Adam, accompanied by a sergeant of Engineers, Sergeant Gasterens, R.E., in command of the wireless telegraphy post, and by the headman of the village. Thanks to their good offices, comfortable shelters were found for us, and I could procure all the food required for the use of my party. The village is of small extent, poor and dreary in appearance. It is said that the sultan Ali Dinar had the greater part of the inhabitants deported a few years ago after confiscating their property, to punish them for showing too much esteem for a certain marabout named Faki Sini, regarded in the district as a worker of miracles. The one that made the deepest impression on the natives, I was assured, consisted in being able to change colour and volume whenever he liked, and even make himself entirely invisible, which did not prevent him from letting himself be surprised and made short work of by the myrmidons of the sultan incensed at his growing prestige.

I had to stay four days in the neighbourhood of Kebkebia, the first part of the time being spent in going back to Rémélé to make arrangements for

the return of my escort and hired camels to Abéché; I also hoped to make the astronomical observations I had been unable to make on the night of my arrival. But I had my labour for my pains. All four days the sky remained almost constantly overcast and the rain fell in torrents, the clouds came in great masses from the west-south-west, and, striking the mountain chain at the foot of which lie Rémélé and Kebkebia, they dissolved in rain that fell at frequent intervals, while on the other side of the chain there fell only rare and insignificant showers.

It was only the last day that I could make the planetary observations required for fixing the positions of Kebkebia, mount Dourboulle, and the summit of the Djebel Marra; this last is notably higher than the 6000 feet above the sea attributed to it by the maps of Africa: my first calculations allowed me to fix its altitude somewhere between 9000 and 9800 feet.

I left Kebkebia on July 2, starting in the afternoon in an easterly direction, skirting the foot of mount Dourboulle on its southern side. The track, cleared of scrub for a width of a dozen yards, lay along a ground rocky indeed, but presenting no serious difficulties. We came across no villages, though the country is inhabited. Here and there on the hillsides one could see stone enclosures, in groups of twenty to thirty, which till a short time previously had been villages whose inhabitants had withdrawn higher up the mountain in order to escape, so at least we were told, from the former sultan's incessant and vexatious requisitions. They were not themselves described to us as particularly desirable, being inclined to banditism; but I can offer no evidence on the question, for they did not trouble the march of my little caravan.

On July 4, for the third and last time, I crossed the line that separates the waters of the Chad basin from that of the Mediterranean, at the Kowra Pass, which is at an altitude of about 4000 feet; then, coming down from spur to spur across the Djebel Kowra I reached the Djebel Om, a very broken region, chaotic in appearance and covered with scanty scrub, stunted, prickly, and almost leafless, where our exhausted camels found but little sustenance. From place to place we crossed recently worked deposits of salt. The salt is very much mixed with earth, and the richest beds are indicated by the swollen, cracked, and friable character of the soil. As in

other salt-producing regions in Central Africa, the salt-bearing earth is washed for a longer or shorter time in washing and filtering baskets; then, when the saline solution has become concentrated enough, it is heated in clay jars, on the inside of which the salt crystallizes as the water evaporates. The product thus obtained, though impure and grey-coloured, is pleasant to the taste, and supplies a great part of the market in Dar Four and the neighbouring countries.

In the afternoon of the 5th, leaving behind us the last salt-beds of Om Bakour, we got clear away from the mountainous zone and made our way for four days across the undulating plains that stretch eastwards beyond El Fasher. The further I went the clearer grew the panorama of the chain I had just crossed. Spur after spur, fantastically shaped, extended in long succession to the north, while towards the west and the south the summits of the Dourboulle and the Djebel Marra towered above the rest of the mountains and stood out boldly against the sky, especially at dusk, a moment at which the light was particularly favourable for the observations required for determining their position and altitude. In the plain of shifting sand, dotted here and there with isolated rocks of huge size, real natural geodetic signals, the landscape stretched away monotonously, almost without trees or even grass. The fertilizing rains of the first few days of July not having reached further than the djebels I had just crossed, the sowing had not begun, and the inhabitants of the villages that succeeded one another at regular intervals down the valleys I traversed were feeling a little uneasy.

At sunrise on July 9, after passing by the hamlet of Zaïdia, I came in sight of the capital of Dar Four; it seemed to be a place of considerable extent, and to consist of thatched huts grouped by distinct quarters along the east side of a bare valley. In the uniform grey of the city I hardly noticed more than one remarkable building, white, and shaped like a tiara, and dominating the northern part of the town; and towards the centre a clump of green trees, from which emerged a construction of European style. The former was the Koumba of Zakaria Zata, the tomb of the sultan Ali Dinar's father; the latter was the sultan's old palace turned into the residence of the Governor of the Province.

Beyond the town I could see low lines of hills, on the north the Djebel Wana, and on the east the Djebel Fasher, at the foot of which a year before the Forian army had been routed by the Anglo-Sudanese troops of Colonel Kelly. To the south a sandy plain of a fine tawny colour stretched away to the horizon, intersected by the long, dark green ribbon of the Wadi El Ko, a sub-tributary through the Bahr el Ghazal of the Nile. Westwards various djebels of greater or less importance stood out in broken lines against the distant curtain of the great chain of western Dar Four. A few moments later I was joined by a group of horsemen: it was His Excellency the Governor of Dar Four, Lieut.-Colonel R. V. Savile Pasha, who bade me welcome and took me to the Residency, where the most cordial hospitality awaited me.

*El Fasher.*—On the evening of my arrival I installed as usual the prismatic astrolabe and the box of chronometers for my daily astronomical observation, and when it was finished I was filled with a deep and intimate joy: after eighteen years of persistent effort I had at last reached the geographical goal that I had set myself to attain in Central Africa. That last observation, made in the palace yard of El Fasher, set the seal, once for all, on the liaison of the geodetic systems of the basins of the Niger, the Chad, and the Nile, for the longitude of El Fasher had just been determined by the officers of the Sudan Survey Department by the aid of the telegraph line recently established between Khartoum and El Fasher. I had to stay twelve days in this town in order to carry out, in conference with the Governor of Dar Four, a mission with which I had been entrusted by the Governor of the Territory of the Chad. This mission concerned the policing of the borderland of the two Governments, and the settlement of the claims arising out of depredations committed by the rebel tribes of Ennedi. After we had come to a complete understanding I drew up, in collaboration with Mr. A. C. Pilkington, a provisional map, on a scale of 1/1,000,000, of the part of the Franco-Anglo-Egyptian borders affected by our agreement. During all this time, need I say that I was the object of the utmost kindness and attention on the part of the Governor and the British officers who surrounded him. Their friendly reception of me remains one of my most treasured recollections of this journey.

El Fasher seemed to be a town of from fifteen to twenty thousand inhabitants, and one of the finest-looking native cities I have seen in Central

Africa; it is built on sand-dunes surrounding a temporary lake that dries up a few weeks after the end of the rainy season, and in which in the dry season the natives dig hundreds of wells, the water of which is then sold at an average price varying between a halfpenny and a penny a gallon. The town stands on two sides of the lake, somewhat in the shape of a circumflex accent, open to the southward, and whose apex is marked, roughly speaking, by the Koubba of Zakaria; the eastern side of this angle is more particularly occupied by traders and natives, while the governor's palace and the greater part of the official buildings are on the western side. Between the business town and the administrative town lies a great square, a sort of Champ de Mars where festivals, parades, and reviews take place, and where once a week the band of the battalion gives a concert.

What struck me most in this town is its well-kept and green appearance; the streets are wide, the houses in good repair and surrounded with trees (mostly serrahs). There are none of the hovels, the broken-down walls, the heaps of refuse so often found in Sudanese cities, except perhaps on the south side, where, at the time of my passing through the town, a group of Fellatas had set up a camp of dirty little straw huts in which men, women, children, and cattle sprawled in an indiscriminate heap.

The sultan Ali Dinar, who had spent part of his youth in the valley of the Nile with the Khalif of the Mahdists, had acquired there a taste for green trees, fine houses, and broad avenues. His palace had been carefully constructed. The principal building, a rectangular white house two stories high, surmounted by a terrace, opened northwards on to a garden planted with palms and lemon-trees. The rooms were large and comfortable, and from the second storey windows the Sultan could see not only the whole of his palace and his capital, but also a vast panorama over the surrounding plain, the valley of the Wadi El Ko, the mountains of Kebkebia, and even the Djebel Marra, whose imposing mass can be seen when the sky is very clear, more than 70 miles to the south-west. Other houses, less sumptuous, but more original because local in style, equally attract one's notice in the interior of this palace, in which one loses one's self in a labyrinth of walls, courtyards, and outbuildings. These houses are large round huts with simple clay walls, but whose roofs, admirably thatched, are often connected by long wide verandahs. These were the apartments of the princesses, light,

roomy, and comfortable. Ali Dinar's æsthetic preoccupations have been rare among Sudanese monarchs, but it must be admitted that in order to embellish his palace and his capital he had all but ruined his kingdom, reducing half the population to a sort of semi-slavery, filling his harem with concubines, distributing his subjects' cattle among his favourites and the Arab merchants who brought him precious merchandise and weapons and ammunition sent by the Senoussists. He dreamed of extending his empire, and lent a too ready ear to the preachers of the Holy War, who, under the ægis of the Grand Senoussi and the Grand Turk, dreamed of driving French and British out of Africa. It was with him as with so many other despots: he fell through pride. Had he shown more wisdom and diplomacy he might well have been reigning still in Dar Four.

There would be many more things to say about El Fasher, but I have already dallied too long over the pleasant memories left me by my sojourn in that town. I beg to be excused inasmuch as, though I was still 1700 miles from Cairo, I considered myself as having reached the end of my journey. There only remained three weeks' march with camels that would bring me to the railway terminus at El Obeid across an inhabited country not merely known but already organized; I must leave the pleasure of describing it to one or another of the British officers who have conquered and pacified it, and who know it better than I, who passed through it too quickly to be able to study it as it deserves.

*From El Fasher to Cairo.*—I left El Fasher in the evening of 21 July 1917, passing through Um Gedada and Dam Gamad to El Nahud, where I arrived on August 4. I left again on the 6th, deeply touched by the hearty welcome of the District Inspector, Major J. G. N. Bardwell. On August 13, towards four in the afternoon, as I came within sight of El Obeid, I heard for the first time in five years the whistle of a locomotive, and its strident note was sweeter to my ears than the most classical music, for it told me that I had at last reached the gate of civilization; and the same evening, at dinner with His Excellency the Governor of Kordofan, Mr. J. W. Sagar, the sight of the graceful and charmingly dressed ladies who were present confirmed that delightful impression.

The next day was a very busy one, for I had to discharge my native escort, pay my camel-drivers, put in order, mend, and bring to the train my numerous cases of instruments, collections, and documents, in order to take on the Wednesday the bi-weekly train. I was only able to do so thanks to the unwearied kindness of the Governor and of the Garrison Commander, Major T. S. Vandeleur, D.S.O.

On August 15, at 7 o'clock in the morning, I took the train for Khartoum. The faithful blacks who had come with me all the way from Borkou were filled with gaping wonder at the sight of the long heavy string of carriages moving by itself. His Excellency the Governor and the Garrison Commander had come to the station to wish me a happy end to my travels, and to see that I had everything I wanted. Let me be allowed here to express once more my lively gratitude!

Then followed two long days in the train across the wide plains of Kordofan, the crossing of the White Nile by a monumental bridge, then the arrival on the Blue Nile at Sennar, where passengers were waiting who had come from the Upper Nile; then Wad Medina in the afternoon, and finally, in the middle of the night, Khartoum.

I stayed a week in Khartoum, where I was the guest of the Civil Secretary, Feilden Pasha, and Dr. P. S. Crispin, Director of the Medical Service. It was an enchanting week that I spent in that pearl of the Sudan, which is already visited by many a tourist, so great was the consideration shown me by my hosts and by the high officials and officers of the capital.

I left Khartoum on August 24, arrived in Cairo in the morning of the 28th, and on the 30th had the honour of being presented at Alexandria by the French Diplomatic Agent to His Excellency the British High Commissioner in Egypt, Sir Reginald Wingate.

As there was no boat ready to start for France, I was able to satisfy my impatience to see an up-to-date fighting front by a visit to the British front lines opposite the Turkish trenches which at that time defended Gaza. Then, returning to Alexandria, I embarked for Malta. From there I reached Syracuse, and thence, by Messina, Naples, Rome, and Modane, I arrived on

1 October 1917 in Paris, and from there a few weeks later I joined the French front.

## 10. Conclusions.

*Geographical Results.*—In the course of this lengthy statement I have set forth in their respective places the principal geographical results obtained during the last five years of my stay in Central Africa; but it will perhaps be convenient to group them in a separate paragraph.

In the first place, the great geographical problem of ancient fluvial communication between the basins of the Chad and the Nile is definitely solved; the mountainous barrier encircles the basin of the Chad from the Toummo Mountains on the north to the Djebel Marra on the south-east, passing through the massif of Tibesti, the plateau of Jef-Jef, the tablelands of Erdi and Ennedi, the hills of Zagawa, and the mountains of western Dar Four.

In the second place, the lowest altitudes of the Chad basin are found in the plains of the low-lying region situated to the north-east of Lake Chad, which we have designated as “the Lowlands of the Chad.” The lowest altitude, of 160 metres (about 520 feet), was found in the ancient lake of Kirri, at a distance of about 250 miles from Lake Chad.

It is towards this low-lying zone that all the great valleys of the hydrographic system of the Western Sahara seem to converge. It is to be presumed that, such being the conditions, the tracing of a hypsometric curve of 250 or 260 metres of altitude (that is to say, slightly superior to that of the actual Chad) would fix the limits, in the region of the Chad, the Lowlands of the Chad, and Borkou, of the ancient Central African lake zone, the existence of which is proved by the agreement of the geological, topographical, ichthyological, malacological, and other observations made in these regions in the course of the last twenty years. Are we to see in the remains of this former Caspian of the Sahara the Chelonide marshes of the geographers of the ancient world? To do so would not be altogether unreasonable if it be taken into account that, so far as I am aware, there is not to be found in the south-west of the Libyan desert any other low-lying

region combining conditions so favourable to the existence of a vast zone of lake or marsh.

Again, if we bear in mind certain local traditions declaring that towards the beginning of the nineteenth century native navigators were able to go in boats from the Chad to the Lowlands of the Chad by the Bahr el Ghazal (an assertion that the present appearance of Lake Kirri, recently dried up, makes sufficiently probable), one may conclude that until the early centuries of the Christian era this low-lying and now completely waterless region of the lowlands of the Chad may have been a great zone of lakes and marshes dotted with sandy or rocky archipelagoes.

Other facts may equally be noted in corroboration of this hypothesis. Firstly, the numerous layers of shells of river molluscs and the large quantity of fish-bones to be met with there: among the latter a fragment of a skull and vertebræ examined by M. J. Pellegrin, which he thought were to be attributed to a Nile perch (*Lates Niloticus*, L.) of about 6 or 7 feet in length (in the *Comptes Rendus de l'Académie des Sciences*, tome 168, No. 19, p. 963. Séance de 12 May 1919); and the discovery of an elephant skeleton in a region where neither grass nor water is any longer to be found. Attention might also be drawn to the rock-drawings of Yarda, where hippopotami are represented among horses, camels, dogs, and ostriches; or to the numerous ruins of settled villages found all up and down, especially where the Bahr el Ghazal falls into the Djourab. Lastly, it may be mentioned that on the platform of certain rocks in Borkou may be found great cemeteries that a native chief attributes to a completely vanished race of "black Christians." But our researches revealed to us no trace or vestige of Christian religion, perhaps because we could not devote enough time to them.

A third important result has been to reveal the geographical form of important mountain masses like Tibesti and Ennedi, hitherto shown in a very imperfect fashion on the maps of Africa, and the existence of another important massif called that of Erdi, connecting the two above mentioned. Moreover, the information we received permits us to reveal to geographers the existence in the centre of the Lybian desert of yet another mountain

mass, the Djebel El Aouinat, situated about 150 miles south-east of the oasis of Koufra, and of which the altitude probably exceeds 4000 feet.

A fourth interesting result has been the precise determination of the difference of longitude Paris-Faya by direct hearing of the wireless time-signals of the Eiffel Tower. Numerous rectifications of the positions attributed to various important points have resulted, the most notable being that which throws more than 50 miles to the N.N.W. the positions attributed by Nachtigal to Bardaï, the peak of Toussidé, the valley of Zouar, etc.

A fifth important result is furnished by the discovery in northern Borkou of the *Harlania Harlani*, which authorizes us to affirm the Upper Silurian age of all the sandstone sedimentary formations of Tibesti, Erdi, and Ennedi.

A sixth point will also, no doubt, be remarked by geographers: from the peak of Toussidé that dominates the north-west of the Tibestian massif to the Djebel Marra overlooking the plains of south-western Dar Four, that is to say, for more than 800 miles in a straight line, numerous hypsometric determinations have been effected which modify—sometimes by several thousand feet—the altitudes of the chief summits of the mountain chain that separates the basin of the Chad from that of the Mediterranean: in Tibesti, Toussidé, 10,700 feet instead of 8200, Emi Koussi, 11,200 feet; in Ennedi, the plateau of Erdébé, 4300 feet; in Tama, the peak of Niéré, 4700 feet; in Dar Four, the peak of Dourboulle, 7200 feet, the Djebel Marra, 9800 feet instead of 6000. These figures are given merely as an indication subject to the rectifications that will follow the revision now proceeding of the summary calculations rapidly effected during my journey.

Lastly, the establishment of the geographical liaison between the Niger, the Chad, and the Nile, by a chain of astronomical positions determined with very satisfactory exactitude, constitutes a seventh result, all the more interesting in that it will permit the drawing up of four sheets of the international map of the world, thanks to the 10,000 kilometres of surveys traced by my collaborators and myself during this long expedition.

From this geographical liaison allow me to pass to another kind of liaison and say a few words on a subject I have particularly at heart, and

which is the conclusion not only of this five years' journey but also of all the journeys I have had the opportunity of making in Central Africa since the beginning of the twentieth century,—I mean the importance, I will even say the necessity, of Franco-British collaboration in the great work of African civilization.

When I first set foot on the Dark Continent, in 1896, tropical and still mysterious Africa was a subject of discussions and rivalries between French and British colonials; but at the present time twenty years of fruitful emulation have ended in a definite and final division of our various possessions, and it seems to me that henceforth Africa is destined to be the tangible pledge of the union of our two countries.

I believe that in England as in France a considerable number of thoughtful men hold that it is above all to the African continent that we must look in a very large proportion for the supply of raw material and foodstuffs that we need. The question is whether it is more to the advantage of France and England to co-operate as closely as possible in developing these vast and practically unworked regions, or whether it is preferable for them to pursue this object separately, each country limiting its means of action to its own sphere of influence.

For my part, I hold that the answer is not doubtful: our two countries should unite their resources for a loyal collaboration in this essential work, so as to assure its complete success as rapidly as possible. I know that the problem is no very simple one; but have we not solved harder ones in the course of these last years, when for both our countries the question was “to be or not to be”? And since it would appear that the great and formidable economic struggle that is beginning on the morrow of the victory is destined to be as keen, if not keener, than the military struggle, it seems to me that the hearty, loyal, and complete union of our efforts can alone assure us of success.

*The Trans-Sudanese.*—It is an axiom henceforth beyond argument that the utilization of the riches running to waste in Tropical Africa cannot be seriously taken in hand until an adequate system of railways is constructed. Allow me, in bringing this lecture to an end, to explain what seems to me

the most rational way of conceiving the general programme of the African railways north of the equator.

In the first place, we must endow Africa with a great transcontinental line from west to east, destined to ensure rapid communication between the different French and British colonies bordering on the Sudan. I have proposed for this railway the name “Transsudanese” (*Comptes Rendus* of the Academy of Sciences, vol. 169, p. 418. Sitting of 1 September 1919 (Gauthier Villars, Paris)); and its main lines, roughly indicated by the natural features of Africa, and following the 13th degree of north latitude, should include the following points:—

- (a) Dakar and Konakry, starting-points on the Atlantic Ocean;
- (b) Ouagadougou, Sokoto, Kano, Fort Lamy, Khartoum, crossing the French Sudan, British Nigeria, the French territory of the Chad, and the Anglo-Egyptian Sudan;
- (c) Port-Sudan and Djibouti, termini on the Red Sea.

Secondly, along this “Transsudanese” would be formed junctions at the most suitable points, with local branch lines from the different French and British colonies that succeed one another along the Atlantic coast from the mouth of the Senegal to that of the Congo.

Thirdly, this railway system would be connected with the Mediterranean ports—on the east by the Nile valley railway from Khartoum to Cairo; on the west by a French “Transsaharian,” starting from the great bend of the Niger and connecting with the railway systems of Tunis, Algeria, and Morocco, and at some future time with that of Europe by a tunnel under the Straits of Gibraltar, or simply by train-ferry.

Among the many reasons urgently in favour of the construction of the Transsudanese, I will confine myself to stating what seems to me the most important and perhaps the least known, the question of labour. For it is generally agreed that the opening up of Tropical Africa cannot be undertaken without the large co-operation of black labour. Now, for long years to come four-fifths of that labour will have to be supplied by the Sudanese populations, much less wild and much less indolent than the great majority of the coast populations, and consequently better fitted to lend

useful aid to European enterprises. This Sudanese population, which may be estimated at some fifteen millions at the lowest count, is spread over more than a million square miles (4000 miles from west to east from the Atlantic to the Red Sea, and 250 to 300 miles from north to south, between the 11th and the 15th degrees of north latitude).

To recruit workmen scattered over such vast distances and convey them without loss of time to the points where European enterprises are ready to employ them, it is evident that an unbroken line of railway must pass through the total length of the inhabited zone—that is to say, of Sudanese Africa. And it is of supreme importance that this railway should not have to take into account the political frontiers of the various colonies passed through, and that its one concern should be to traverse the regions in which the population is densest.

Such is one of the main considerations that fix the choice of the itinerary and bring me to the conclusion that the Transsudanese—a work of general interest in Africa, and more particularly a work of specially Franco-British interest—ought to be undertaken without delay, and pushed forward as actively as may be by the cordial co-operation of France and Great Britain.

These remarks do not apply to the local railways of the different colonies, though they may be expected to participate largely in the traffic of the Transsudanese, either by carrying down the products of the interior to the ports of the coast or by giving access to the regions in need of development, and in which Sudanese labour will be required. I am of opinion that these railways, limited as they are to the particular territories of the several colonies whose economic development they ensure, should continue to be constructed and managed, as hitherto, by the colonies they serve: those colonies should bear the expense of such local lines by their own financial resources, or by those placed at their disposal by the mother-country.

As for the Transsaharian, destined to connect the railways of North Africa (Morocco, Algeria, Tunis) with those of the Niger basin, I have had the opportunity of saying in another place that it has become a vital necessity of French colonial policy in Africa—a necessity that the great war

has proved to demonstration. For this reason I hold that its construction should be regarded as a work of strictly national interest. Still, a glance at the map will convince the observer of the profit that will accrue to the British West African colonies, especially when it becomes possible to cross from Europe to Africa without the inconvenience of a sea-passage. I have often been met by the objection that the Transsaharian “will not pay”; that it will be almost exclusively a strategic railway, very laborious to construct, and very costly to keep in working order. Such is not my opinion. The Transsaharian, once the junction effected with the Transsudanese, will connect two exceedingly rich regions—the Africa of the Arab and Berber races and Black Africa. Between these regions a considerable commercial traffic will arise, which will have an influence as great or even greater than that of the Transsudanese itself on the economic development of Africa; its receipts per kilometre will be as large if not larger than those of the most favoured of the railways running from the colonies along the coast inland towards the Sudan, for the Transsaharian will be the direct means of penetration into the richest regions of tropical Africa, not only from North Africa, but also from the whole of Western Europe.

## 1871-1919

May I say one word about Tibesti and Borkou, and so conclude? Half a century ago, when Nachtigal, after exploring the Tibesti, came to the shores of Lake Chad, before setting out again to complete his work by the exploration of Kanem and Borkou, he learnt by letters from Tripoli the victories that his native country of Germany had won over France. And again, when he returned to Europe after four long years of absence, he found that peace had been made two years earlier, and that our provinces of Alsace and Lorraine had become part of Germany and were called the Reichsland; France, humiliated, was just finishing the payment to the conqueror of the millions that were to hasten the liberation of her territory.

By a striking example of the way in which history sometimes repeats itself, but with a difference, war was once more forced on France by Germany at a moment when French explorers had just set foot in Borkou and Tibesti in order to rectify, revise, and complete the unfinished work of the German explorer! And the joy that filled the heart of Nachtigal when he returned to Europe to find his country triumphant, and her borders widened with the spoils of war, swells in our hearts to-day! For it is Germany now that knows the humiliation of paying millions to obtain the liberation of her own territory, while the tricolour floats over Metz and Strasburg, and watch indeed is kept, but to other music, on the Rhine!

From this parallel, may I venture to conclude that in her treasure-house of colonial jewels France may well find a place for arid Borkou and the barren Tibesti. For would it not seem that they are, in some sort, talismans, and that when Gaul and German grapple on the banks of the great river that was set by nature and destiny to hold them apart, Fortune, that wayward goddess, shall give victory to whichever country has a son exiled in those mysterious regions, seeking, by rock and desert, new ways across their ancient sand?

*[Translated from the French by W. G. Tweedale, M.A., Oxon.]*

Before the paper the PRESIDENT said: It is a special pleasure to us to welcome here this evening that well-known French explorer and geographer, Colonel Tilho. We had been long hoping to have the pleasure of receiving him and of hearing an account of his recent journeys from 1912 to 1917, but owing to the press of official business he was not able to come here in the summer, and it is only by the greatest good fortune, and by the exercise of a little tactful pressure upon the different Governments, that he has been able to be present this evening. This is not the first occasion upon which he has been before the Society. He gave us a most interesting paper about ten years ago, so that he is not a stranger, and we are very glad to welcome him again. What he will describe to us this evening will be his journeys in Central Africa and the French Sudan between the years 1912 and 1917; and it was for the valuable work which he did during those journeys and for his general contribution to geographical knowledge that we awarded him, two years ago, our Patron's Gold Medal. I have, therefore, very great pleasure in introducing Colonel Tilho to you and asking him now to address us.

*Colonel Tilho then gave in French a summary of the paper printed above, and a discussion followed.*

The PRESIDENT (after the paper): Sir Henry McMahon, who was High Commissioner in Egypt during part of the war, is present here, and we shall be very glad if he will kindly make some observations in regard to Colonel Tilho's interesting lecture.

Colonel Sir HENRY McMAHON: We are much indebted to Colonel Tilho for a most interesting paper to-night. It is not only of very great interest, but a valuable contribution to geographical knowledge. I will leave the discussion of the lecture as regards its geographical and cartographical aspect to others, but there is one portion of the paper to which I should like to call your attention. As Colonel Tilho has told you, during the war the Germans and Turks got a footing in Tripoli. He has told you how Enver Pasha's brother, Nuri Bey, landed on that coast, and with him many Germans. Their object was to get into touch with the Senussi; raise the whole country against us through the Senussi influence, and threaten our

western flank both in Egypt and the Sudan. They very nearly succeeded; and if our brave allies, the French, had not forestalled them in the country described to-night, they would undoubtedly have established themselves there. It is a valuable objective as being the first place in which water and supplies can be got after leaving the oasis of Kufra. We will imagine for one moment that they had established themselves there. You can at once see what a dangerous focus of intrigue and unrest, what a source of danger it would have been on our flank all along our western front. Having forestalled the enemy there, no further trouble ensued, but our friend the Sultan of Darfur, who misjudged the time of the Senussi arrival and counted too confidently on their aid, had already started hostilities with us, and a war ensued which in times of peace would have attracted wide public attention but in the days when our interest was so concentrated on other fronts it almost escaped notice. Suffice to say that by a brilliant series of military operations, our troops, under the direction of Sir Reginald Wingate, the Sirdar of the Sudan, drove him out of his capital and took the whole of his country. If the Senussi had at this time been established with their German and Turkish assistants on our flank, it might have been a very different job indeed. I look upon this incident as an object lesson of the good that co-operation can effect in a work of this kind, and it is, I hope, not only an object lesson of what has been done in the past times of war, but an augury of what we can do and should do between us in the future times of peace. As Colonel Tilho has explained to you, co-operation is essential for the development of this great country of Africa, and I trust that it will be the guiding principle of our two great nations not only in the development of that country, but in furthering the welfare of the backward peoples placed under our guardianship.

The PRESIDENT: The French Military Attaché is present and we should be very pleased if he would kindly address us.

General the VISCOMTE DE LA PANOUSE: Je ne savais pas que j'aurais à prendre la parole ce soir en sorte que je me trouve un peu pris au dépourvu. Je vous demanderais donc la permission de m'exprimer en Français. Il y a quelques vingt ans, il eut été impossible de discuter ici dans une atmosphère de calme et de confiance mutuelle une question relative au centre du Continent Africain. Heureusement depuis cette époque, grâce aux

bienfaisants accords de 1904, les malentendus entre le Royaume Uni et la France se sont dissipés, l'Entente Cordiale est née, elle s'est développée et elle a vu son couronnement dans une alliance militaire étroite et loyale pendant la plus grande guerre que le monde ait vue. Le Colonel Tilho vous a exposé pourquoi dans le développement économique de ce Grand Centre Africain, l'action unie des deux grandes Nations est nécessaire sous peine d'aboutir à un gaspillage inutile d'efforts et d'argent. Mais je vois aussi une autre raison pour laquelle nous devons travailler ensemble; l'Empire Britannique et la France ont lutté pendant cette grande guerre pour faire triompher les principes du droit et de la liberté contre l'oppression et la barbarie. Notre victoire nous a créé des obligations et en particulier celle de défendre les populations noires contre la tyrannie des marchands d'esclaves et de l'oppression des sectes musulmanes et de leur donner le bien-être auquel a droit tout être humain. Ce devoir ne sera utilement rempli que si nos nations s'entendent sur les mesures à prendre et les réalisent en commun. La belle œuvre d'humanité à accomplir sera ainsi un nouveau lien entre les deux Grandes Puissances qui se partagent le continent Africain.

The PRESIDENT: We have been fortunate to catch Sir Harry Johnston. He is one of our greatest authorities upon Africa generally, both Central and Northern. We should be very glad if he would make some remarks.

Sir HARRY JOHNSTON: I had the honour some years ago, just after the war had started, of showing you a somewhat similar map of Africa with railways designed on it partly by my own fancy, and I may say to a great extent by following French fancies too; for about that time I had been in the north of Africa, and had been allowed to pursue for a certain distance the tracing of the projected trans-Saharan railway, the progress of which was only stopped by the war. I conceived then the idea that it was of the highest importance to Western Europe that that line should be made, though I, like most of you, did not appreciate the influence on affairs that the submarine would have; but of course that conviction has been strengthened by the events of the war. Had we had the trans-Saharan railway in existence during the war we should not have suffered as much as we did from the loss of some of the most important materials for our industries caused by the interruptions of the sea routes, the destruction of steamers, etc. It is a matter of absolute necessity, I consider, that that trans-Saharan line should be made

to link up the valley of the Niger with French North Africa, and further with Western Europe; because, as Colonel Tilho has pointed out, the channel between Tangier and the Spanish coast could be easily patrolled and kept free of submarines, and even crossed by train ferries. Then another point I should like to raise is as to the further exploration of those Tibesti highlands and the lofty plateaus that are connected with them on the north-west and south-east. Colonel Tilho did not mention in his discourse what he said to me privately, that he had found in some parts of that region, possibly Borku, fossilized bones of elephants. He has referred to the native legends and to the drawings on the rocks which point to the existence of hippopotami in regions now entirely devoid of surface water. He showed some of these engravings. They are very similar to rock drawings which can be traced right across the Sahara desert, exhibiting a fauna now completely passed away. One reason why Tibesti should be explored is, that we might find there the fossil and semi-fossil remains of a very extensive tropical African fauna, because that isthmus of high land between the south of Tunis on the north, and Darfur and the regions round Lake Chad on the south, seems to have been the principal route by which the fauna of Miocene and Pliocene Europe and the Mediterranean basin reached Tropical Africa. There are more and more indications that the Sahara desert to the west and the Libyan and Nubian deserts to the east were formerly under water, and therefore checked the progress of beasts and man across the Sahara into Central Africa; but this high ridge always remained well above the limits of such lakes, marshes, or inland seas. Tibesti was a well-watered region with at one time quite a heavy rainfall down to about twenty thousand years ago.

Before the war suspended such enterprises, the savants of France were exploring the wonderful sub-fossil remains of Algeria which revealed to us the existence there of a mammalian fauna resembling that of modern tropical Africa, of the region south of the Sahara. With that fauna were mingled in a very interesting degree creatures which at the present time are restricted to India. For instance, there was something so like an Indian elephant that it might be called the Indian elephant, existing almost down to the human period in Algeria. There was a wild camel, an equine resembling a zebra; there were gnus, hartebeests, oryxes, and other types of modern African antelopes; and there was a Tragelaph allied to the Nilghai; there

was a huge buffalo with almost incredible horns—14 feet long—incredible were it not that its existence is proved not only by its fossil remains but by the drawings of primitive man. The Foureau-Lamy Expedition, I believe, found many of the dry torrent-beds of the elevated Ahaggar region choked with hippopotamus bones. There is everything to point to quite a recent and rapid change in the climate of the Sahara, which, well within the human period, was a region abounding in water derived from a heavy rainfall, and richly endowed with forest areas, as we may see from the remains of petrified trees. This will bring home to you what gains might come to science and to our knowledge of the evolution of life on this planet if we could only thoroughly explore the Sahara, and above all such regions as the Tibesti highlands.

Major HANNS VISCHER: Just after I had crossed the Sahara, some years ago, I had the great pleasure to meet Colonel Tilho in Nigeria; and last time we met—I think in 1909—to celebrate our homecoming in Paris, we spoke of the work in Africa of our two respective countries. During my journey, and whenever I met the French in those regions, I was particularly impressed by the difficulties and privations these officers suffered so cheerfully. In Nigeria we had our railway, and we got frequent leave. As I remembered those isolated posts in the heart of the Sahara, while looking at the pictures we saw to-night, separated by hundreds of miles, rarely getting a mail or any provisions from the coast during those long years of war, when few boats went to the West Coast of Africa, I was filled with admiration for the work done by Colonel Tilho and his comrades. In the course of his lecture the Colonel showed clearly how necessary it is for us to co-operate in Africa, not only for the welfare of the native people but also for the very existence of our respective colonies. He has shown to us to-night how well we can complement each other. When that German-Turkish column advanced south across the desert, at a moment when we had sent most of our troops from Nigeria to East Africa, it would have been a hard thing for the people in our colony if the officers under Colonel Tilho's orders, assisted by some native troops sent north from Nigeria, had not been able to arrest the enemy's progress.

The PRESIDENT: I know you will all want me to congratulate Colonel Tilho on your behalf on the lucid, graceful, and humorous lecture he has

given us this evening. There has been great talk about the co-operation between us and the French, and I think we might go a little deeper even than that. When we can get a French officer like Colonel Tilho over here in the flesh, and can hear from his own lips what he has done, when he shows us pictures of the kind of country he has had to make his way through, the kind of people he has had to make friends with: when we see all that, certainly we who have had to do similar work in other parts of the world—and probably you at home, even though you have not had that great pleasure and honour, must have a very deep fellow-feeling with him and his compatriots—we feel that there is something deep and common between us when we realize so vividly the work that they are doing, the difficulties that they have had to encounter, and the great work of civilization and humanization which they are carrying on in these far remote recesses of Central Africa. We have had to do the same things ourselves in other parts of the world. We see the results of our own efforts, and Colonel Tilho this evening has shown us what the French have done in opening out the great arid wastes of the Sahara desert and the French Sudan. What they have done and what we have done is good for the world as a whole. It has all been opened out gradually in the course of years, not only for the French and not only for the British, but for all nations. Therefore we here in England, we in this Society, will send forth a very hearty word of congratulation to the French, and especially to Colonel Tilho, for the great work which they are doing in Central Africa. He has made very important geographical discoveries, and has referred to new methods of geographical observation. Wireless telegraphy for the purpose of determining longitude is a comparatively new method, but one which is vastly valuable, because, as we who have tried to determine longitudes in far-away places know, in old days it was impossible to get the longitude at all exactly. We could get the latitude fairly accurately, within a few hundred yards, but longitude we could never get to within a few miles. Now by means of wireless telegraphy we are able to get longitude with almost complete exactitude, even in the heart of the French Sudan. Colonel Tilho has also made a slight allusion to another modern invention which I think in future will prove of great service, and that is the aeroplane. We shall hear more of that at our next meeting; but when you see those vast waterless regions, when you hear from Colonel Tilho of the enormous difficulty in getting across them with

camels, then we see of what use the aeroplane might have been made for preliminary geographical reconnaissance. Those two inventions, I am certain, will be of enormous service to geography. I now wish on your behalf to tender to Colonel Tilho a most hearty vote of thanks for his lecture this evening, and also for his great kindness, at considerable personal inconvenience, in coming across from Paris to give us this paper.

## FOOTNOTES:

- [1] A sort of camp-followers whose business in life is warfare in all its branches except that of fighting: experts in all manner of desert craft, scouts, flank-guards, finders of strayed camels or sorely needed wells. Swift to detect the incompetence or bad faith of local guides, they form the necessary complement to the fighting strength of any expedition in Central Africa.
- [2] This account will be published in the next number of the *Journal*.—ED. G.J.

\*\*\* END OF THE PROJECT GUTENBERG EBOOK THE  
EXPLORATION OF TIBESTI, ERDI, BORKOU, AND ENNEDI IN 1912-  
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